

An Informational Parsimony Perspective on Probabilistic Symmetries

Hippolyte Charvin

Nicola Catenacci Volpi

Daniel Polani

Adaptive Systems Research Group, University of Hertfordshire

H.CHARVIN@HERTS.AC.UK

N.CATENACCI-VOLPI@HERTS.AC.UK

D.POLANI@HERTS.AC.UK

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Abstract

Extraction of structure, in particular of group symmetries, is increasingly crucial to understanding and building intelligent models. In particular, some information-theoretic models of parsimonious learning have been argued to induce invariance extraction. Here, we formalise these arguments from a group-theoretic perspective. We then extend them to the study of more general probabilistic symmetries, through compressions preserving geometric measures of complexity. More precisely, our framework implements a trade-off between compression and preservation of the divergence from a given hierarchical model, yielding a novel generalisation of the Information Bottleneck framework. Through appropriate choices of hierarchical models, we fully characterise (in the discrete and full support case) channel invariance, channel equivariance and distribution invariance under permutation. Allowing imperfect divergence preservation then leads to principled definitions of “soft symmetries”, where the “coarseness” corresponds to the degree of compression of the system. In simple synthetic experiments, we demonstrate that our method successively recovers, at increasingly compressed “resolutions”, nested but increasingly perturbed equivariances, where new equivariances emerge at bifurcation points of the trade-off parameter. Our framework suggests a new path for the extraction of generalised probabilistic symmetries.

Keywords: Probabilistic symmetries, Information Bottleneck, Geometric Complexity

1. Introduction

Group symmetries have become highly relevant to the study of intelligence, from the structure of sensory processing (Keller et al., 2024) and its relation to an agent’s actions (Godon et al., 2020; Keurti et al., 2024) to equivariant neural networks (Gerken et al., 2023) and structure-discovering AI models (van der Ouderaa et al., 2024). This relevance is often understood as a consequence of the pervasiveness of symmetries in the natural world: biological and artificial systems interacting with such a highly structured environment should *leverage* its symmetries, e.g., by internalising them into their own information-processing. But what leverage do symmetries provide exactly? Intuitively, the presence of symmetries in a system allows for a *simpler* description of it — thus potentially improving the efficiency of learning and generalisation about this system. In other words, a system’s symmetries afford the possibility of *informationally parsimonious* descriptions of it.

More precisely, the *projection on orbits* of a symmetry group’s action can be seen as an information-preserving compression, as it preserves the information about anything invariant

under the group action. This suggests that projections on orbits might be solutions to well-chosen rate-distortion problems, hence opening the way to the integration of group symmetries into an information-theoretic framework. If successful, such an integration could formalise the link between symmetry and information parsimony, but also (i) yield natural ways to “soften” group symmetries into flexible concepts more relevant to real-world data — which often lacks exact symmetries despite exhibiting a strong “structure” — and (ii) enable symmetry discovery through the optimisation of information-theoretic quantities.

As a first step in this direction, we introduce a novel rate-distortion-inspired framework whose solutions capture, or at least mimic, the projection on orbits of certain group symmetries. We call it Divergence Information Bottleneck (DIB), as it generalises the Information Bottleneck (IB) method (Tishby et al., 2000). Here, the encoders trade-off compressing data with preserving its divergence from a given hierarchical model, potentially under constraints on the encoder shape. As divergences from hierarchical models are geometric measures of complexity (Ay et al., 2011), this setting produces *complexity-preserving* compressions. With appropriate choices of hierarchical models and shape constraints, we obtain encoders which, for full divergence preservation, characterise various group-theoretic symmetries. Allowing only *partial* divergence preservation then leads to principled definitions of *soft symmetries*, whose “softness” is parametrised by the compression-divergence preservation trade-off: i.e., softer symmetries correspond to more compression and less preserved divergence.

The IB method has previously been argued to extract invariances (Achille and Soatto, 2018). In Section 2, we prove that it indeed characterises group-theoretic channel invariances. This motivates the DIB framework, which we specialise to characterise the equivariances of a channel $p(Y|X)$ and the invariances of a distribution $p(A)$ (Section 3). We then present simple synthetic numerical experiments on soft equivariances (Section 4), where we study channels satisfying a series of nested equivariances that have been perturbed to various degrees. We show that our framework recovers the perturbed equivariances, at successive bifurcation points of the trade-off parameter corresponding to increasingly compressed resolutions. Finally, we present the limitations of our approach (Section 5), and conclude in Section 6.

Assumptions and notations All alphabets are finite, except bottleneck alphabets $\mathcal{T} := \mathbb{N}$. The probability simplex defined by an alphabet $\mathcal{A}$ is denoted by $\Delta_{\mathcal{A}}$. The set of conditional probabilities, also called channels, from $\mathcal{A}$ to $\mathcal{B}$, resp. to $\mathcal{A}$ itself, is denoted by $C(\mathcal{A}, \mathcal{B})$, resp. $C(\mathcal{A})$. Functions are seen as deterministic channels. Channels are often seen as linear maps between vector spaces of measures, where the image of the distribution p through the channel γ is written $\gamma \cdot p$. By extension, $f \cdot a := f(a)$ for an element a and a function f . The symbol $\circ$ denotes channel composition. The set of bijections of $\mathcal{A}$ is $\text{Bij}(\mathcal{A})$, the uniform distribution $\mathcal{U}(\mathcal{A})$, the identity map $e_{\mathcal{A}}$, and $\mathcal{S}^c := \mathcal{A} \setminus \mathcal{S}$ for $\mathcal{S} \subseteq \mathcal{A}$. For $p_1 \in \Delta_{\mathcal{A}}, p_2 \in \Delta_{\mathcal{B}}$, their *tensor product* is defined through $(p_1 \otimes p_2)(a, b) := p_1(a)p_2(b)$. Similarly $(\mu \otimes \eta)(b, b'|a, a') := \mu(b|a)\eta(b'|a')$ for $\mu \in C(\mathcal{A}, \mathcal{B}), \eta \in C(\mathcal{A}', \mathcal{B}')$.

2. Information Bottleneck and Group Invariances

The IB implements a trade-off between compressing a variable, say X , and preserving the information that X carries about a second variable Y . More precisely, let $p(X, Y) \in \Delta_{\mathcal{X} \times \mathcal{Y}}$

with $p(X)$ full-support. The corresponding IB problem is (Gilad-Bachrach et al., 2003)

$$\text{IB}(\lambda) := \arg \min_{\substack{q(T|X) \in \mathcal{C}(\mathcal{X}, \mathcal{T}) : \\ I_q(T; Y) \geq \lambda}} I_q(X; T), \quad (1)$$

where $q(X, Y, T) := p(X, Y)q(T|X)$ and $0 \leq \lambda \leq \Lambda := I(X; Y)$.

On the other hand, the *channel invariance group* G_{ci} of $p(Y|X)$ is the group of $\sigma \in \text{Bij}(\mathcal{X})$ such that $p(Y|X) \circ \sigma = p(Y|X)$, with projection on orbits written $\pi_{\text{ci}} : \mathcal{X} \rightarrow \mathcal{X}/G_{\text{ci}}$. Crucially, here π_{ci} characterises G_{ci} : it can be easily verified that $\sigma \in G_{\text{ci}} \Leftrightarrow \pi_{\text{ci}} \circ \sigma = \pi_{\text{ci}}$.

We now want to show that the solutions $\kappa \in \text{IB}(\Lambda)$ essentially coincide with π_{ci} , thus yielding an information-theoretic characterisation of G_{ci} through such κ . We will need the equivalence relation

$$x \sim_{\mathcal{X}} x' \Leftrightarrow p(Y|x) = p(Y|x'), \quad (2)$$

with corresponding partition $\bar{\mathcal{X}}$ and projection $\pi_{\mathcal{X}} : \mathcal{X} \rightarrow \bar{\mathcal{X}}$; along with the following notion.

Definition 1 *The set of congruent channels (Ay et al., 2017) from $\mathcal{A}$ to $\mathcal{B}$, denoted by $C_{\text{cong}}(\mathcal{A}, \mathcal{B})$, is that of channels γ such that there exists a function $f : \mathcal{B} \rightarrow \mathcal{A}$ with $f \circ \gamma = e_{\mathcal{A}}$.*

In particular, composing an encoder κ with a congruent channel γ can be seen as a trivial operation, in that the output of κ can be unambiguously recovered from that of $\gamma \circ \kappa$.

Theorem 2 *For $\Lambda := I(X; Y)$ and all $\sigma \in \text{Bij}(\mathcal{X})$, the following holds:*

- (i) $\text{IB}(\Lambda) = \{\gamma \circ \pi_{\mathcal{X}} : \gamma \in C_{\text{cong}}(\bar{\mathcal{X}}, \mathcal{T})\}$.
- (ii) Let $\kappa \in \text{IB}(\Lambda)$. Then $\sigma \in G_{\text{ci}}$ if and only if $\kappa \circ \sigma = \kappa$.
- (iii) If $\sigma \in G_{\text{ci}}$, then $\kappa \circ \sigma = \kappa$ also holds for all $0 \leq \lambda \leq \Lambda$ and $\kappa \in \text{IB}(\lambda)$.
- (iv) The projection $\pi_{\mathcal{X}}$ defined by $\sim_{\mathcal{X}}$ coincides with the projection on orbits π_{ci} .

Proof See Appendices A and B. ■

Crucially, point (ii) means that invariances are those bijections σ such that the effect of transforming $\mathcal{X}$ with σ is “quotiented out” by $\kappa \in \text{IB}(\Lambda)$. Point (i) explains why: these κ implement precisely (up to trivial transformations) the quotient of $\mathcal{X}$ by the equivalence relation $\sim_{\mathcal{X}}$ that equates elements of $\mathcal{X}$ providing the same information about Y (see (2)). Point (iii) shows that the “quotienting out” of invariances by bottlenecks also occurs for all values of the trade-off parameter λ , even though it is only a full characterisation for $\lambda = I(X; Y)$. Point (iv), combined with point (i), means that the projection π_{ci} , defined purely in group-theoretic terms, is characterised as the solution to the zero-distortion case of a generalised rate-distortion problem, here the IB (Zaidi et al., 2020). Note that point (i) is redundant with existing results (Shamir et al., 2010);¹ and point (ii) is not surprising as previous work already linked the IB method to invariance extraction (Achille and Soatto, 2018). However, our group-theoretic formalisation provides guidance for generalisations of this phenomenon: i.e., for reformulating and softening probabilistic symmetries with the language of information theory. The following sections provide first steps in this direction.

1. Point (i) can be seen as the fact that $\text{IB}(\Lambda)$ consists of minimal sufficient statistics of X w.r.t. Y , proven in (Shamir et al., 2010). But our new proof also yields point (iii) and mirrors that of Theorem 3 below.

3. Divergence Information Bottleneck and Group Symmetries

3.1. General framework

Fix a distribution $p = p(A) \in \Delta_{\mathcal{A}}$, an exponential family $\mathcal{E} \subseteq \Delta_{\mathcal{A}}$, and a subset of encoders $C \subseteq C(\mathcal{A}, \mathcal{T})$. We then define the *Divergence Information Bottleneck* (DIB) as

$$\text{DIB}(\lambda) := \arg \min_{\substack{\kappa \in C \\ D(\kappa \cdot p || \kappa \cdot \mathcal{E}) \geq \lambda}} I_{\kappa}(A; T), \quad (3)$$

where $0 \leq \lambda \leq \Lambda := D(p || \mathcal{E})$, and, denoting by $\text{cl } \mathcal{E}$ the topological closure of $\mathcal{E}$ in $\Delta_{\mathcal{A}}$,

$$D(p || \mathcal{E}) := \inf_{r \in \text{cl } \mathcal{E}} D(p(A) || r(A)) = D(p(A) || \tilde{p}(A)), \quad (4)$$

$$D(\kappa \cdot p || \kappa \cdot \mathcal{E}) := \inf_{r \in \text{cl } \mathcal{E}} D((\kappa \cdot p)(T) || (\kappa \cdot r)(T)) = D((\kappa \cdot p)(T) || (\kappa \cdot \tilde{p})(T)). \quad (5)$$

Here $\tilde{p} \in \text{cl } \mathcal{E}$ is the unique distribution which simultaneously achieves the minimum in (4) and (5) (see Appendix C.1). While $D(p || \mathcal{E})$ is the divergence of p from $\mathcal{E}$, here $D(\kappa \cdot p || \kappa \cdot \mathcal{E})$ measures the “divergence of p from $\mathcal{E}$ in the latent space $\mathcal{T}$, through the lens of the channel κ ”. Solutions to (3) can thus be seen as optimal compressions of A under the constraint of (partially or fully) preserving the divergence of $p(A)$ from the exponential family $\mathcal{E}$. The choice of C allows to potentially enforce constraints on the shape of encoder channels κ .

Intuitively, $D(p || \mathcal{E})$ measures the presence of a specific structure in $p(A)$, formalised as the divergence from the family $\mathcal{E}$ of distributions which do not have such structure. E.g., for $\mathcal{A} = \mathcal{X} \times \mathcal{Y}$ and $\mathcal{E} := \Delta_{\mathcal{X}} \otimes \Delta_{\mathcal{Y}}$, we have $D(p(X, Y) || \mathcal{E}) = I(X; Y)$: the corresponding DIB (with e.g. $C = C(\mathcal{X} \times \mathcal{Y}, \mathcal{T})$) is a *mutual information-preserving* joint compression of X and Y (Charvin et al., 2023). More generally, the divergence from a *hierarchical model* $\mathcal{E}$ measures the complexity of a system’s given set of interdependencies (Ay et al., 2011). The DIB is tailored for this setting, where solutions to (3) are thus *complexity-preserving optimal compressions*. However, Theorem 3 below holds for any exponential family $\mathcal{E}$.

Let us define, on $\mathcal{A}$, the equivalence relation

$$a \sim a' \Leftrightarrow p(a)\tilde{p}(a') = p(a')\tilde{p}(a), \quad (6)$$

with corresponding partition $\bar{\mathcal{A}} := \{\mathcal{A}_j\}_{j=1, \dots, n}$ and projection $\pi : \mathcal{A} \rightarrow \bar{\mathcal{A}}$.

Theorem 3 *If $C = C(\mathcal{A}, \mathcal{T})$ and $\text{supp}(p(A)) = \mathcal{A}$, then $\text{DIB}(\Lambda) = \{\gamma \circ \pi : \gamma \in C_{\text{cong}}(\bar{\mathcal{A}}, \mathcal{T})\}$.*

I.e., for full support $p(A)$ and no constraints on the shape of encoders, the fully divergence-preserving solutions $\kappa \in \text{DIB}(\Lambda)$ coincide, up to trivial transformations, with the clustering of $\mathcal{A}$ defined by the relation (6). The theorem is proven in Appendices A and C.2.

3.2. Application to equivariances

Consider now $\mathcal{A} = \mathcal{X} \times \mathcal{Y}$ equipped with a full support distribution $p(X, Y)$; in particular, $p(Y|X)$ is well-defined. The group G_{ce} of *channel equivariances* is the group of pairs $(\sigma, \tau) \in \text{Bij}(\mathcal{X}) \times \text{Bij}(\mathcal{Y})$ such that $p(Y|X) \circ \sigma = \tau \circ p(Y|X)$. Crucially, as was the case for invariances (see Section 2), the projection on orbits π_{ce} characterises the group G_{ce} : it can be easily verified that $(\sigma, \tau) \in G_{\text{ce}} \Leftrightarrow \pi_{\text{ce}} \circ (\sigma \otimes \tau) = \pi_{\text{ce}}$.

We want to design a DIB problem that mimics this property of π_{ce} . Note that here π_{ce} does not compress $\mathcal{X}$ and $\mathcal{Y}$ separately but *jointly* (Charvin et al., 2023), so that it is natural to impose no constraint on encoders' shape: $C = C(\mathcal{X} \times \mathcal{Y}, \mathcal{T})$. Moreover, it can be verified² that $\pi_{ce}(x, y) = \pi_{ce}(x', y')$ implies $p(y|x) = p(y'|x')$. Based on this observation, we search for an exponential family $\mathcal{E}$ such that the relation $\sim$ from equation (6) becomes

$$(x, y) \sim (x', y') \Leftrightarrow p(y|x) = p(y'|x'). \quad (7)$$

This is achieved by choosing

$$\mathcal{E} = \mathcal{E}_{ce} := \{r(X)\mathcal{U}(\mathcal{Y}), r(X) \in \Delta_{\mathcal{X}}\},$$

which yields $\tilde{p}(X, Y) = p(X)\mathcal{U}(\mathcal{Y})$, so that the relation $p(x, y)\tilde{p}(x', y') = p(x', y')\tilde{p}(x, y)$ (see equation (6)) here means $p(y|x) = p(y'|x')$. Note that $\mathcal{E}_{ce}$ coincides with the hierarchical model of probabilities on $\mathcal{X} \times \mathcal{Y}$ that actually depend only on $\mathcal{X}$. Borrowing from the geometric approach to complexity (Ay et al., 2011), we thus interpret $D(p(X, Y) || \mathcal{E}_{ce})$ as the “degree to which the system (X, Y) is more than just X ”, or equivalently, “the amount of information that $p(Y|X)$ carries about (X, Y) ”. More precisely, $\mathcal{E}_{ce}$ is the family of probabilities $r(X, Y)$ such that $r(Y|x)$ (when well-defined) is always equal to the maximum entropy distribution $\mathcal{U}(\mathcal{Y})$. This condition means, intuitively, that the channel $r(Y|X)$ is “maximally uninformative”, i.e., that it “carries no information” about the system (X, Y) . Through the decomposition $r(X, Y) = r(X)r(Y|X)$, this is equivalent to the intuition that all the information about (X, Y) is contained in the marginal $r(X)$.

The divergence $D(p || \mathcal{E}_{ce})$ then quantifies “how far” $p = p(X, Y)$ is from such distributions. Thus the set $\text{DIB}_{ce}(\lambda)$ of solutions to the DIB problem (3) with $\mathcal{A} = \mathcal{X} \times \mathcal{Y}$, $\mathcal{E} = \mathcal{E}_{ce}$ and parameter λ is that of optimal compressions *preserving the information carried by the channel $p(Y|X)$* (to the degree λ).

Theorem 4 *The following holds, for $\Lambda := D(p || \mathcal{E}_{ce})$ and all $(\sigma, \tau) \in \text{Bij}(\mathcal{X}) \times \text{Bij}(\mathcal{Y})$:*³

- (i) *Let $\kappa \in \text{DIB}_{ce}(\Lambda)$. Then $(\sigma, \tau) \in G_{ce}$ if and only if $\kappa \circ (\sigma \otimes \tau) = \kappa$.*
- (ii) *If $(\sigma, \tau) \in G_{ce}$, then $\kappa \circ (\sigma \otimes \tau) = \kappa$ also holds for all $0 \leq \lambda \leq \Lambda$ and $\kappa \in \text{DIB}_{ce}(\lambda)$.*
- (iii) *The projection π defined by $\sim$ in equation (7) does not, in general, coincide with π_{ce} .*

See Appendix C.3 for the proof, which relies on Theorem 3. Most importantly, point (i) means that equivariances of $p(Y|X)$ are those pairs of transformations (σ, τ) such that the effect of simultaneously transforming $\mathcal{X}$ with σ and $\mathcal{Y}$ with τ is “quotiented out” by the coarse-grainings $\kappa \in \text{DIB}(\Lambda)$, making these transformations indiscernible from the identity. In this sense, the DIB_{ce} framework provides an *information-theoretic characterisation of equivariances*. It can be summarised with the following commutative diagrams:

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{p(Y|X)} & \mathcal{Y} \\ \sigma \downarrow & & \downarrow \tau \\ \mathcal{X} & \xrightarrow{p(Y|X)} & \mathcal{Y} \end{array} \Leftrightarrow \begin{array}{ccc} \mathcal{X} \times \mathcal{Y} & \xrightarrow{\sigma \otimes \tau} & \mathcal{X} \times \mathcal{Y} \\ & \searrow \kappa & \swarrow \kappa \\ & \mathcal{T} & \end{array}$$

2. See Lemma 15 in (Charvin et al., 2023).

3. Appendix C.4 clarifies how this work relates to our previous results in (Charvin et al., 2023).

which means that the commutativity of the left-hand-side diagram — i.e., the defining property of equivariances — is equivalent to that of the right-hand-side one — an information-theoretic property. Moreover, point (ii) in Theorem 4 means that the “quotienting out” of equivariances happens actually for all granularities λ , even though the equivalence only holds for $\lambda = \Lambda$. Point (iii) says that, unfortunately, the clustering π obtained from $\text{DIB}_{\text{ce}}(\Lambda)$ does not always coincide with the projection on orbits π_{ce} (see Appendix C.3 for more details).

We can now draw upon our new information parsimony perspective on equivariances to soften this group-theoretic notion, where each granularity λ defines a corresponding set of soft equivariances. Let $0 \leq \lambda \leq \Lambda$. We define a λ -*equivariance* as a pair $(\mu, \eta) \in C(\mathcal{X}) \otimes C(\mathcal{Y})$ such that there exists $\kappa \in \text{DIB}_{\text{ce}}(\lambda)$ with $\kappa \circ (\mu \otimes \eta) = \kappa$. In other words, a soft equivariance is defined through the very same equation $\kappa \circ (\sigma \otimes \tau) = \kappa$ that characterises exact equivariances, but where the fully information-preserving compression κ is now only a *partially* information-preserving compression. Moreover, we allow μ and η to be non-invertible and stochastic.

To conclude this section, let us point out that the classic IB can be recovered as a DIB with the same exponential family $\mathcal{E}_{\text{ce}}$ as for equivariances, but with shape constraints $C \subsetneq C(\mathcal{A}, \mathcal{T})$ which impose that κ can only compress $\mathcal{X}$ and not $\mathcal{Y}$. See Appendix C.5.

3.3. Application to distribution invariances

We now proceed similarly for transformations leaving a given distribution invariant. I.e., let $p \in \Delta_{\mathcal{A}}$ be full support, and define the group G_{di} of *distribution invariances* as the group of $\Phi \in \text{Bij}(\mathcal{A})$ such that $p(\Phi \cdot A) = p(A)$. As we do not consider any structure on $\mathcal{A}$, we choose unconstrained encoders, i.e., $C = C(\mathcal{A}, \mathcal{T})$. Moreover, as $\Phi \in G_{\text{di}}$ if and only if $p(a) = p(\Phi \cdot a)$ for all a , it is natural to search for an exponential family yielding the equivalence relation $a \sim a' \Leftrightarrow p(a) = p(a')$. It can be easily verified that this is achieved by choosing only the uniform distribution: $\mathcal{E} = \mathcal{E}_{\text{di}} := \{\mathcal{U}(\mathcal{A})\}$. Intuitively, here the DIB problem, which we denote by DIB_{di} , preserves (partially or wholly) the divergence $D(p(A) || \mathcal{U}(\mathcal{A}))$ of $p(A)$ from the uniform distribution: i.e., it preserves the “degree to which $p(A)$ is deterministic”.

Theorem 5 *The following holds, for $\Lambda := D(p || \mathcal{E}_{\text{di}})$ and all $\Phi \in \text{Bij}(\mathcal{A})$:*

- (i) *Let $\kappa \in \text{DIB}_{\text{di}}(\Lambda)$. Then $\Phi \in G_{\text{di}}$ if and only if $\kappa \circ \Phi = \kappa$.*
- (ii) *If $\Phi \in G_{\text{di}}$, then $\kappa \circ \Phi = \kappa$ also holds for all $0 \leq \lambda \leq \Lambda$ and $\kappa \in \text{DIB}_{\text{di}}(\lambda)$.*
- (iii) *The projection π defined by $\sim$ coincides with the projection on orbits π_{di} .*

The proof relies on Theorem 3 (see Appendix C.6). Interpretations of points (i) and (ii) are analogous to those for equivariances. Point (iii) highlights that here, π_{di} and π do coincide. Eventually, one can directly adapt the definition of soft equivariances to one for soft distribution invariances.

3.4. Relevant computational and conceptual tools

Here, we present an algorithm approximating solutions to the DIB (3) for unconstrained encoders, and two relevant notions. Consider the Lagrangian relaxation of (3),

$$\arg \min_{\kappa \in C} \left[I_{\kappa}(A; T) - \beta D(\kappa \cdot p || \kappa \cdot \tilde{p}) \right], \quad (8)$$

where $\beta \geq 0$. For $C = C(\mathcal{A}, \mathcal{T})$, deriving the Lagrangian above w.r.t. κ yields a fixed-point equation that all local minimisers must satisfy (similarly to the classic IB). From this fixed-point equation, we obtain a Blahut-Arimoto-like (BA) algorithm with the same guarantees as BA for the classic IB (Tishby et al., 2000) (see Appendices D.1 and D.2).⁴ In the following, we will write κ_β for the output of the BA algorithm (i.e., a local minimiser) with parameter β , and also $I_\beta := I_{\kappa_\beta}(A; T)$ and $D_\beta := D(\kappa_\beta \cdot p || \kappa_\beta \cdot \tilde{p})$. Note that both I_β and D_β increase with β .

Now the *effective cardinality* (Zaslavsky and Tishby, 2019) of some $\kappa \in \text{DIB}(\lambda)$ is defined as the minimum number of symbols t necessary to describe the output of κ (see Appendix D.3 for a formal definition). In all our numerical experiments, we observed that: (i) similarly as for the classic IB, effective cardinality monotonically increases with β , and (ii) changes of effective cardinality coincide with discontinuities in the slope of the curve $\beta \mapsto (I_\beta, D_\beta)$, which is reminiscent of the second-order bifurcations observed for the IB (Zaslavsky and Tishby, 2019). We will thus here refer to changes of effective cardinalities as bifurcations.

Eventually, we want to investigate whether the equation $\kappa_\beta \circ \Phi = \kappa_\beta$ is satisfied for varying β and varying $\Phi \in G$, with G some fixed subgroup of $\text{Bij}(\mathcal{A})$. But numerically, it is also important, when this equation is not exactly satisfied, to quantify the extent of the deviation. We propose to use the divergence defined for all channel $\kappa \in C(\mathcal{A}, \mathcal{T})$ as

$$D^p(\kappa || C_G) := \min_{\nu \in C_G} D^p(\kappa || \nu) := \min_{\nu \in C_G} \sum_{a \in \text{supp}(p(A))} p(a) D(\kappa(T|a) || \nu(T|a)),$$

where $C_G := \{\nu : \forall \Phi \in G, \nu \circ \Phi = \nu\}$ is the family of channels that are exactly input-symmetric w.r.t G . Intuitively, $D^p(\kappa || C_G)$ measures the divergence of the channel κ from being input-symmetric for the action of G on the distribution $p(A)$. In particular, $D^p(\kappa || C_G) = 0$ if and only if $\kappa \circ \Phi = \kappa$ for all $\Phi \in G$. See Appendix D.4 for more details.

4. Numerical experiments

4.1. Synthetic numerical experiments on equivariances

The concept of soft λ -equivariance from Section 3.2 is motivated by the case of full information preservation $\lambda = \Lambda$, where our new definition of Λ -equivariances coincides with that of classic group equivariances. However, it is not a priori clear that, once $\lambda < \Lambda$, our generalisation is consistent with the intuition of approximate symmetry. Here, we provide a sanity check in this direction, in a simple synthetic grid-world scenario. We start from an exactly equivariant channel $p(Y|X)$, which we synthetically perturb in such a way to render some of its equivariances more perturbed than others. We then expect that (i) all the perturbed equivariances should be recovered as soft λ -equivariances by the DIB_{ce} framework, once the reduction of the parameter λ enforces sufficient compression, and (ii) more perturbed equivariances should need more compression before being recovered as soft equivariances — because it means that more noise needs to be “overlooked”.

We compute the DIB_{ce} solutions with the BA algorithm described in Appendix D.2, combined with reverse deterministic annealing, starting from $T = (X, Y)$ for large β (similarly

4. The convergence speed (and thus computational feasibility) of this and other BA algorithms is difficult to estimate — particularly as they exhibit critical slowing down near bifurcations (Agmon et al., 2021).

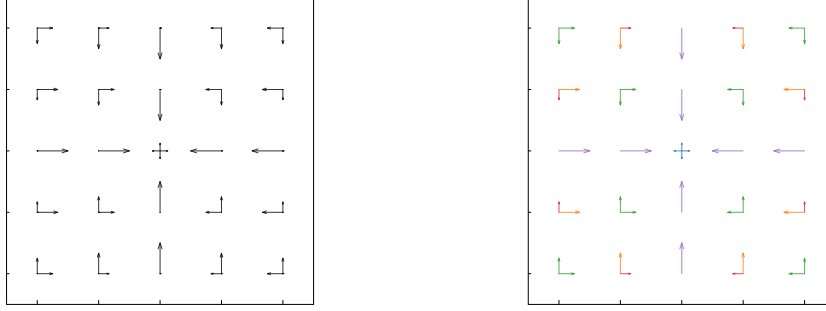


Figure 1: Left: representation of $p(Y|X)$, where X is the position on the grid, Y the gradient direction, and probabilities are proportional to arrow lengths. Thus equivariances are here pairs (σ, τ) that send each arrow on an arrow of equal length. Right: same figure with colors representing a clustering of $\text{supp}(p(X, Y))$ — which defines a clustering of $\mathcal{X} \times \mathcal{Y}$ if we add the cluster $\text{supp}(p(X, Y))^c$, made of position-orientation pairs with probability 0, i.e., with no arrow. The latter clustering is obtained in 2 distinct ways: (i) as the projection on orbits of the equivariance group of $p(Y|X)$, and (ii) as the clustering defined by relation (7).

as for the classic IB in (Zaslavsky and Tishby, 2019)). All our experiments were simulated on a PC having 32GB of RAM and a 2.3GHz 12th generation i7 CPU.

Here, $\mathcal{X}$ stands for positions on a 5×5 grid, and $\mathcal{Y}$ for a gradient with 4 possible directions. Thus $p(Y|X)$ describes the probability of a direction at a given position, which can be thought of, e.g., as a nutrient gradient sensed by a bacteria. We choose uniform $p(X)$ (choosing non-uniform $p(X)$ resulted in similar results). As seen in Figure 1, left, $p(Y|X)$ has many symmetries: it can be verified that the equivariance group of $p(Y|X)$ has 6 distinct orbits (one is $\text{supp}(p(X, Y))^c$), represented in Figure 1, right. Moreover, even though we saw in Theorem 4, point (iii), that the projection on orbits π_{ce} and the clustering π defined by relation (7) do not generally coincide, here they do coincide. Thus Figure 1, right, also represents π .

From Section 3.4, we have $D^p(\kappa_\beta || C_{G_{ce}}) = 0$ if and only if $\kappa_\beta \circ (\sigma, \tau) = \kappa_\beta$ for all $(\sigma, \tau) \in G_{ce}$. Theorem 4, point (ii), suggests that this equation may indeed hold for all β .⁵ As a sanity check, we thus computed the DIB_{ce} bottlenecks κ_β for $0 \leq D_\beta \leq \Lambda$, and indeed obtained $D^p(\kappa_\beta || C_{G_{ce}}) \leq 3 \cdot 10^{-16}$ for all β . We also noted that the bottlenecks’ effective cardinality monotonically increases from 1 for $D_\beta = 0$ to 6 for $D_\beta = \Lambda$.

We then perturb $p(Y|X)$ with two random perturbations. The first one, of larger amplitude, breaks some equivariances in G_{ce} , but not all of them. More precisely, after the perturbation, $p(Y|X)$ still satisfies the equivariances from its subgroup $G_{rota} \subsetneq G_{ce}$ generated by rotating both the positions and the gradient directions by 90 degrees. The second perturbation applied to $p(Y|X)$, of smaller amplitude, breaks all the remaining equivariances from G_{rota} . We thus obtain a new $p(Y|X)$ which, intuitively, is still “approximately”

5. Here, the full support assumption, which is required in Theorem 4, does not hold for $p(X, Y)$. We leave to future work the theoretical study of the case $\text{supp}(p(X, Y)) \subsetneq \mathcal{X} \times \mathcal{Y}$.

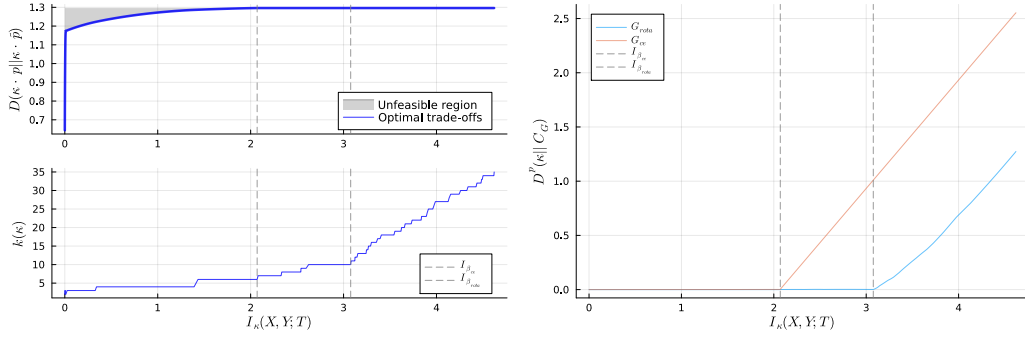


Figure 2: $D_\beta := D(\kappa_\beta \cdot p \| \kappa_\beta \cdot \tilde{p})$ as a function of $I_\beta := I_{\kappa_\beta}(X, Y; T)$. Bottom left: Effective cardinality $k(\kappa)$ as a function of I_β . Right: Divergence of compression channels κ_β as a function of I_β , for the groups G_{ce} and G_{rota} . The vertical dashed lines represent specific bifurcations of the parameter β at which $D^p(\kappa_\beta \| C_{G_{\text{rota}}})$, resp. $D^p(\kappa_\beta \| C_{G_{\text{ce}}})$, approximately vanishes (in decreasing order of I_β).

equivariant, but where the approximate equivariances in $G_{\text{ce}} \setminus G_{\text{rota}}$ are *coarser* than those in G_{rota} , because the perturbation was larger for the former than for the latter.

We compute 1000 DIB_{ce}-bottlenecks for varying β (which took 339 seconds). The resulting information curve (I_β, D_β) , along with the corresponding effective cardinalities, are shown in Figure 2, left. As for the classic IB, we obtain a non-decreasing and concave information curve, and an increasing effective cardinality (except for small I_β , which could be due to numerical errors).

Crucially, we then observe (Figure 2, right) that for decreasing β , the divergences $D^p(\kappa_\beta \| C_{G_{\text{rota}}})$ and $D^p(\kappa_\beta \| C_{G_{\text{ce}}})$ successively vanish, at bifurcation values β_{rota} , resp. $\beta_{\text{ce}} < \beta_{\text{rota}}$. Thus the perturbed equivariances are here recovered by the DIB_{ce} method as *soft* equivariances, for large enough compression. Moreover, as the equivariances from G_{rota} have been less perturbed than those in the remaining of G_{ce} , here the degree of compression required to recover an approximate equivariance scales with the “coarseness” of that equivariance.

Eventually, note that, in Figure 2, left, the gain in divergence D_β from $I_\beta = I_{\beta_{\text{ce}}}$ to the maximum value I_{max} of I_β is negligible, whereas $I_{\text{max}} - I_{\beta_{\text{ce}}}$ is large. This resonates with the intuition that underlying symmetries in raw data afford a potentially drastic informational compression (I_β here), under a negligible loss in informational accuracy (D_β here).

5. Limitations

Our core results are of theoretical nature, and hold in the discrete and full support case. At this stage, it is still unclear whether and how they extend to continuous and non fully supported distributions. Numerically, the BA class of algorithms addresses only the discrete case and generally scales unfavourably in larger scenarios. Future work could make the DIB problem amenable to deep network optimisation by adapting the classic IB’s variational bounds (Alemi et al., 2017).

Moreover, the DIB corresponding to equivariances does not yield the projection on orbits under the group’s action (Theorem 4, point (iii)). Our DIB_{ce} framework still characterises group equivariances for maximum parameter $\lambda = \Lambda$, but for $\lambda < \Lambda$, this limitation might make the concept of λ -equivariance ill-adapted to the kind of “soft equivariances” that we hope to identify in real-world data. Future work should test further the scientific relevance of λ -equivariances as defined here, and potentially design a new instance or variation of the DIB which would capture the projection on orbits of the equivariance group. The link between our soft equivariances and other proposals of generalised equivariances (Wang et al., 2022; Romero and Lohit, 2022; Song et al., 2023; Ashman et al., 2024) should also be clarified.

Let us stress that at the current stage, our method provides an information-preserving compression κ corresponding to — or mimicking — the symmetry group’s projection on orbits, but does not discover the symmetries themselves. However, our “projection on orbits”-oriented perspective grounds group symmetries in IB theory, which could serve as a building block for information theory-based symmetry discovery.

6. Conclusion

Motivated by the ability of the classic IB to implicitly extract channel invariances, we investigated generalizations of this phenomenon. For this, we introduce the Divergence IB, a novel and substantial generalization of the classic IB. We show how this method can generalize the informational characterisation of invariances to that of channel equivariances and distribution invariances. Crucially, expressing these symmetries through IB-like trade-offs yields a natural softening of these very stringent group-theoretic notions. This suggests a principled route to extract the data’s underlying symmetries through *complexity-preserving coarse-grainings* of it, thus exposing the data’s “platonic core”, so to say.

However, while we only investigated some canonical examples of symmetries, the DIB framework is highly versatile. With other exponential families $\mathcal{E}$ and channel shape constraints $\mathcal{C}$, this method could help discover novel kinds of scientifically relevant structures.

Eventually, our work suggests a new path to formalise the intuition that informationally parsimonious systems, e.g., embodied agents, should leverage the coarse symmetries underlying the interaction with their environment.

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Appendix A. General notations, definitions and results

The proofs of Theorem 2 (see Appendix B) and Theorem 3 (see Appendix C.2.) are very similar. Here, we collect definitions and pieces of reasoning that are common to both.

We fix a finite set $\mathcal{A}$, a probability $p \in \Delta_{\mathcal{A}}$ which we assume full support, and a channel $\kappa = q(T|A) \in C(\mathcal{A}, \mathcal{T})$. We also consider a partition $\bar{\mathcal{A}} = \{\mathcal{A}_j\}_{j=1,\dots,n}$ of $\mathcal{A}$, and denote by $\pi : \mathcal{A} \rightarrow \bar{\mathcal{A}}$ the corresponding projection. Whenever it can simplify notations, we will identify $\bar{\mathcal{A}}$ with $\{1, \dots, n\}$. We associate to $\kappa = q(T|A)$ a corresponding channel $\bar{\kappa} = \bar{q}(T|\mathcal{A}_j) \in C(\bar{\mathcal{A}}, \mathcal{T})$ defined through

$$\bar{q}(t|j) := \bar{q}(t|\mathcal{A}_j) := q(t|\mathcal{A}_j) := \frac{\sum_{a \in \mathcal{A}_j} q(t|a)p(a)}{p(\mathcal{A}_j)}. \quad (9)$$

Intuitively, $\bar{q}(T|\mathcal{A}_j)$ is the “channel induced by $q(T|A)$ (and $p(A)$) when quotienting its input space $\mathcal{A}$ into the partition $\bar{\mathcal{A}}$ ”. We also define a channel $\kappa_{\pi} = q_{\pi}(T|A) \in C(\mathcal{A}, \mathcal{T})$ through, for all $a \in \mathcal{A}$, $t \in \mathcal{T}$,

$$q_{\pi}(t|a) := (\bar{\kappa} \circ \pi)(t|a) = \sum_j q(t|\mathcal{A}_j) \delta_{a \in \mathcal{A}_j}. \quad (10)$$

Intuitively, $q_\pi(T|A)$ is the “enforced factorisation of $q(T|A)$ through π ”. Indeed: (i) it is a channel defined from $q(T|A)$ that factorises through π ; and (ii), whenever $q(T|A)$ itself factorises through π , then we must have $q(T|A) = q_\pi(T|A)$ (see point (i) in Lemma 8).

Lemma 6 *We have*

$$(i) \quad q_\pi(T) = q(T).$$

$$(ii) \quad I_{q_\pi}(A; T) \leq I_q(A; T), \text{ where equality holds if and only if } q_\pi(T|A) = q(T|A).$$

Proof (i). For all $t \in \mathcal{T}$,

$$\begin{aligned} q_\pi(t) &= \sum_{a \in \mathcal{A}} p(a) q_\pi(t|a) \\ &= \sum_j \sum_{a \in \mathcal{A}_j} p(a) q_\pi(t|a) \\ &= \sum_j \sum_{a \in \mathcal{A}_j} p(a) q(t|\mathcal{A}_j) \\ &= \sum_j p(\mathcal{A}_j) q(t|\mathcal{A}_j) \\ &= \sum_j \sum_{a \in \mathcal{A}_j} q(t|a) p(a) = q(t), \end{aligned}$$

where the second and last equalities use the fact that $\{\mathcal{A}_j\}_j$ is a partition of $\mathcal{A}$, and the fourth uses the definition of $q(t|\mathcal{A}_j)$ in (9).

(ii). We have

$$\begin{aligned} I_q(A; T) &= \sum_{a \in \mathcal{A}, t \in \text{supp}(q(T))} p(a) q(t|a) \log \left(\frac{q(t|a)}{q(t)} \right) \\ &= \sum_{t \in \text{supp}(q(T))} \sum_j \sum_{a \in \mathcal{A}_j} p(a) q(t|a) \log \left(\frac{q(t|a)}{q(t)} \right). \end{aligned}$$

But from the log-sum inequality (Csiszár and Körner, 2011), for fixed $t \in \text{supp}(q(T))$ and fixed j ,

$$\begin{aligned} \sum_{a \in \mathcal{A}_j} p(a) q(t|a) \log \left(\frac{q(t|a)}{q(t)} \right) &= \sum_{a \in \mathcal{A}_j} p(a) q(t|a) \log \left(\frac{p(a) q(t|a)}{p(a) q(t)} \right) \\ &\geq \left(\sum_{a \in \mathcal{A}_j} p(a) q(t|a) \right) \log \left(\frac{\sum_{a \in \mathcal{A}_j} p(a) q(t|a)}{\sum_{a \in \mathcal{A}_j} p(a) q(t)} \right) \\ &= p(\mathcal{A}_j) q(t|\mathcal{A}_j) \log \left(\frac{q(t|\mathcal{A}_j)}{q(t)} \right), \end{aligned} \tag{11}$$

with equality in (11) if and only if $\frac{q(t|a)p(a)}{q(t)p(a)}$ is constant for $a \in \mathcal{A}_j$, i.e., if and only if $q(t|a)$ is constant for $a \in \mathcal{A}_j$. Note that the last line of (11) can be rewritten

$$\begin{aligned} p(\mathcal{A}_j)q(t|\mathcal{A}_j) \log \left(\frac{q(t|\mathcal{A}_j)}{q(t)} \right) &= \sum_{a \in \mathcal{A}_j} p(a)q_\pi(t|a) \log \left(\frac{q_\pi(t|a)}{q(t)} \right) \\ &= \sum_{a \in \mathcal{A}_j} p(a)q_\pi(t|a) \log \left(\frac{q_\pi(t|a)}{q_\pi(t)} \right), \end{aligned}$$

where the second equality uses point (i) proven above. Thus, summing (11) over j and $t \in \text{supp}(q(T))$, we get

$$\begin{aligned} I_q(A; T) &\geq \sum_{t \in \text{supp}(q(T))} \sum_j \sum_{a \in \mathcal{A}_j} p(a)q_\pi(t|a) \log \left(\frac{q_\pi(t|a)}{q_\pi(t)} \right) \\ &= I_{q_\pi}(A; T), \end{aligned}$$

with equality if and only if for all $t \in \text{supp}(q(T))$ and all j , the quantity $q(t|a)$ is constant for $a \in \mathcal{A}_j$ — which means more precisely that for all $a \in \mathcal{A}_j$, we have $q(t|a) = q(t|\mathcal{A}_j)$. In other words, there is equality if and only if $q(t|a) = q_\pi(t|a)$ for all $a \in \mathcal{A}$ and $t \in \text{supp}(q(T))$. As the definition of $q_\pi(T|A)$ clearly implies $\text{supp}(q(T)) = \text{supp}(q_\pi(T))$, the latter is equivalent to $q(T|A) = q_\pi(T|A)$. $\blacksquare$

We now introduce, for all $t \in \mathcal{T}$, the set

$$\mathcal{A}_t^q := \{a \in \mathcal{A} : q(t|a) > 0\}, \quad (12)$$

which can be seen as the “probabilistic pre-image of t through the channel $q(T|A)$ ”.

Lemma 7 *The following are equivalent:*

- (i) *The channel $\bar{q}(T|\mathcal{A}_J) \in C(\bar{\mathcal{A}}, \mathcal{T})$ defined in (9) is congruent.*
- (ii) *For all $t \in \text{supp}(q(T))$, there exists a partition element $\mathcal{A}_j \in \bar{\mathcal{A}}$ such that $\mathcal{A}_t^q \subseteq \mathcal{A}_j$.*

Note that the $\mathcal{A}_j$ satisfying $\mathcal{A}_t^q \subseteq \mathcal{A}_j$ is actually unique, because $\{\mathcal{A}_j\}_j$ is a partition of $\mathcal{A}$, and $\mathcal{A}_t^q \neq \emptyset$ for $t \in \text{supp}(q(T))$.

Proof For any function $f : \mathcal{T} \rightarrow \{1, \dots, n\}$ and all j, j' ,

$$(f \circ \bar{q})(j'|j) = \sum_{t \in \mathcal{T}} \delta_{j'=f(t)} \bar{q}(t|j) = \sum_{t \in \mathcal{T}} \delta_{j'=f(t)} q(t|\mathcal{A}_j) = q(f^{-1}(j')|\mathcal{A}_j),$$

where $f^{-1}(j')$ is the pre-image of j' through f . Thus

$$(f \circ \bar{q})(j'|j) = \delta_{j'=j} \Leftrightarrow q(f^{-1}(j')|\mathcal{A}_j) = \delta_{j'=j} \quad (13)$$

$$\Leftrightarrow q(f^{-1}(j')|\mathcal{A}_j) > 0 \text{ only if } j' = j \quad (14)$$

$$\Leftrightarrow \forall t \in \text{supp}(q(T)), \quad q(t|\mathcal{A}_j) > 0 \text{ only if } f(t) = j \quad (15)$$

$$\Leftrightarrow \forall t \in \text{supp}(q(T)), \quad (\exists a \in \mathcal{A}_j : q(t|a) > 0) \text{ only if } f(t) = j \quad (16)$$

$$\Leftrightarrow \forall t \in \text{supp}(q(T)), \quad (\mathcal{A}_t^q \cap \mathcal{A}_j \neq \emptyset) \text{ only if } f(t) = j \quad (17)$$

where line (14) uses the fact that $q(\cdot|\mathcal{A}_j)$ is a probability measure and $\{f^{-1}(j)\}_j$ a partition of $\mathcal{T}$; line (15) that $q(f^{-1}(j')|\mathcal{A}_j) = \sum_{t:f(t)=j'} q(t|\mathcal{A}_j)$; line (16) that $q(t|\mathcal{A}_j) = \frac{1}{p(\mathcal{A}_j)} \sum_{a \in \mathcal{A}_j} q(t|a)p(a)$ with $\mathcal{A}_j \subseteq \mathcal{A}$; and line (17) the definition of $\mathcal{A}_t^q$. Therefore,

$$\exists f : \mathcal{T} \rightarrow \{1, \dots, n\}, \forall j, j', (f \circ \bar{q})(j'|j) = \delta_{j'=j} \quad (18)$$

$$\Leftrightarrow \exists f : \mathcal{T} \rightarrow \{1, \dots, n\}, \forall j, \forall t \in \text{supp}(q(T)), (\mathcal{A}_t^q \cap \mathcal{A}_j \neq \emptyset) \text{ only if } f(t) = j$$

$$\Leftrightarrow \exists f : \mathcal{T} \rightarrow \{1, \dots, n\}, \forall t \in \text{supp}(q(T)), \mathcal{A}_t^q \subseteq \mathcal{A}_{f(t)} \quad (19)$$

But on the one hand, the statement (18) is the definition of $\bar{q}(T|\mathcal{A}_J)$ being a congruent channel. On the other hand, because any function on $\text{supp}(q(T))$ can be arbitrarily extended to the whole $\mathcal{T}$, the statement (19) is equivalent to

$$\exists f : \text{supp}(q(T)) \rightarrow \{1, \dots, n\}, \forall t \in \text{supp}(q(T)), \mathcal{A}_t^q \subseteq \mathcal{A}_{f(t)},$$

which is clearly a reformulation of point (ii). $\blacksquare$

Lemma 8 Fix a channel $\gamma \in C(\bar{\mathcal{A}}, \mathcal{T})$, and define $\kappa := q(T|A) := \gamma \circ \pi$. Then

(i) γ coincides with the channel $\bar{\kappa} := \bar{q}(T|\mathcal{A}_J)$ defined in equation (9).

(ii) If moreover γ is congruent, then $I_q(A; T) = H(\pi(A))$.

Proof (i). For all $\mathcal{A}_j \in \bar{\mathcal{A}}$,

$$\bar{q}(t|\mathcal{A}_j) := \sum_{a \in \mathcal{A}_j} \frac{p(a)q(t|a)}{p(\mathcal{A}_j)} = \sum_{a \in \mathcal{A}_j} \frac{p(a)\gamma(t|\pi(a))}{p(\mathcal{A}_j)} = \sum_{a \in \mathcal{A}_j} \frac{p(a)\gamma(t|\mathcal{A}_j)}{p(\mathcal{A}_j)} = \gamma(t|\mathcal{A}_j).$$

(ii). Fix a deterministic function $f : \mathcal{T} \rightarrow \bar{\mathcal{A}}$ such that $f \circ \gamma = e_{\bar{\mathcal{A}}}$. Then

$$I_q(A; T) \geq I_q(A; f(T)) = I_q(A; f \circ \gamma(\pi(A))) = I(A; \pi(A)) = H(\pi(A)),$$

where the last equality holds because π is deterministic. On the other hand, as a direct consequence of the factorisation $q(T|A) = \gamma \circ \pi$, we have the Markov chain $A - \pi(A) - T$. Thus $I_q(A; T) \leq I(A; \pi(A)) = H(\pi(A))$. $\blacksquare$

Appendix B. Proof of Theorem 2

The following sections present the successive steps of the proof. Let us first note that all the definitions and statements from Appendix A can be applied here, with $\mathcal{A} = \mathcal{X}$ and $p(A) = p(X)$ (we assumed that $p(A)$ is full support in Appendix A, but also that $p(X)$ is full support in Section 2). We write here $\bar{\mathcal{A}} = \bar{\mathcal{X}} = \{\mathcal{X}_j\}_{j=1, \dots, n}$ and $\pi = \pi_{\mathcal{X}} : \mathcal{X} \rightarrow \bar{\mathcal{X}}$ (these notations coincide with those defined in Section 2); also $q(T|A) = q(T|X)$, $\bar{q}(T|\mathcal{A}_J) = \bar{q}(T|\mathcal{X}_J)$, $q_{\pi}(T|A) = q_{\pi_{\mathcal{X}}}(T|X)$. Moreover, recall that here $q(T|X)$ defines not only a joint distribution $q(X, T)$, but also, together with $p(X, Y)$, a joint distribution $q(X, Y, T)$ through the assumed Markov chain $T - X - Y$. Similarly, $q_{\pi}(T|X)$ defines a joint distribution $q_{\pi}(X, Y, T)$.

B.1. Factorisation for all parameter λ

Proposition 9 *For all $0 \leq \lambda \leq \Lambda$, every solution $q(T|X) \in IB(\lambda)$ factorises as $q(T|X) = \bar{q}(T|\mathcal{X}_J) \circ \pi_{\mathcal{X}}$.*

The proof will derive from the following lemma:

Lemma 10 *For every $q(T|X)$, we have:*

- (i) $q_{\pi_{\mathcal{X}}}(Y; T) = q(Y, T)$, so that in particular $I_{q_{\pi_{\mathcal{X}}}}(Y; T) = I_q(Y; T)$.
- (ii) $I_{q_{\pi_{\mathcal{X}}}}(X; T) \leq I_q(X; T)$, where equality holds if and only if $q_{\pi_{\mathcal{X}}}(T|X) = q(T|X)$.

Proof (i). For all $y \in \mathcal{Y}, t \in \mathcal{T}$,

$$\begin{aligned}
 q_{\pi_{\mathcal{X}}}(y, t) &= \sum_{x \in \mathcal{X}} p(y, x) q_{\pi_{\mathcal{X}}}(t|x) = \sum_{j=1}^n \sum_{x \in \mathcal{X}_j} p(y, x) q_{\pi_{\mathcal{X}}}(t|x) \\
 &= \sum_{j=1}^n \sum_{x, x' \in \mathcal{X}_j} p(y, x) p(x') \frac{q(t|x')}{p(\mathcal{X}_j)} \\
 &= \sum_{j=1}^n \sum_{x, x' \in \mathcal{X}_j} p(y, x') p(x) \frac{q(t|x')}{p(\mathcal{X}_j)} \\
 &= \sum_{j=1}^n \sum_{x' \in \mathcal{X}_j} p(y, x') q(t|x') \\
 &= \sum_{x \in \mathcal{X}} p(y, x') q(t|x') = q(y, t),
 \end{aligned} \tag{20}$$

where the first and last lines use that $\{\mathcal{X}_j\}_j$ is a partition of $\mathcal{X}$; and line (20) uses the definition of the sets $\mathcal{X}_j$ through the equivalence relation $\sim_{\mathcal{X}}$ (see equation (2)): i.e., for $x, x' \in \mathcal{X}_j$, we have for all $y \in \mathcal{Y}$, $p(y|x) = p(y|x')$, which is equivalent to $p(y, x)p(x') = p(y, x')p(x)$.

(ii). We apply point (ii) in Lemma 6. ■

But Lemma 10 means that for the IB problem (1), if we replace the channel $q(T|X) \in C(\mathcal{X}, \mathcal{T})$ by the corresponding $q_{\pi_{\mathcal{X}}}(T|X)$, then (i) the value for of the constraint function is unchanged, and (ii) the value of the target function does not increase, with equality if and only if $q_{\pi_{\mathcal{X}}}(T|X) = q(T|X)$. In particular, if $q(T|X)$ solves the IB problem, then we must have $q(T|X) = q_{\pi_{\mathcal{X}}}(T|X)$: i.e., $q(T|X) = \bar{q} \circ \pi_{\mathcal{X}}$.

B.2. Explicit form of solutions for $\lambda = \Lambda$ (point (i) in Theorem 2)

In this section, we prove point (i) in Theorem 2, i.e., that $IB(\Lambda) = \{\gamma \circ \pi_{\mathcal{X}} : \gamma \in C_{\text{cong}}(\mathcal{X}_J, \mathcal{T})\}$.

We will here denote by $\mathcal{X}_t^q$ the “probabilistic pre-image” $\mathcal{A}_t^q$ from Appendix A (see equation (12)): i.e., for a channel $q = q(T|X)$ and all $t \in \mathcal{T}$,

$$\mathcal{X}_t^q := \{x \in \mathcal{X} : q(t|x) > 0\}. \tag{21}$$

Lemma 11 *We have $I_q(T; Y) \leq \Lambda := I(X; Y)$, and the following are equivalent:*

- (i) $I_q(T; Y) = \Lambda$.
- (ii) *For all $t \in \text{supp}(q(T))$, there exists a partition element $\mathcal{X}_j \in \bar{\mathcal{X}}$ such that $\mathcal{X}_t^q \subseteq \mathcal{X}_j$.*
- (iii) *The channel $\bar{q}(T|\mathcal{X}_J) \in C(\bar{\mathcal{X}}, \mathcal{T})$ defined in (9) is congruent.*

The equivalence of (i) and (ii) means that the constraint $I_q(T; Y) = I(X; Y)$ holds if and only if $p(Y|x)$ is constant on the pre-image $\mathcal{X}_t^q$ of every symbol t .

Proof (i) $\Leftrightarrow$ (ii). We have

$$\begin{aligned} I_q(T; Y) &= \sum_{y \in \text{supp}(p(Y)), t \in \text{supp}(q(T))} p(y)q(t|y) \log \left(\frac{q(t|y)}{q(t)} \right) \\ &= \sum_{y \in \text{supp}(p(Y)), t \in \text{supp}(q(T))} p(y) \left(\sum_x p(x|y)q(t|x) \right) \log \left(\frac{\sum_x p(x|y)q(t|x)}{\sum_x p(x)q(t|x)} \right). \end{aligned}$$

But for all $y \in \text{supp}(p(Y))$ and $t \in \text{supp}(q(T))$, from the log-sum inequality (Csiszár and Körner, 2011), with the convention $0 \log(\frac{0}{0}) := 0$,

$$\left(\sum_x p(x|y)q(t|x) \right) \log \left(\frac{\sum_x p(x|y)q(t|x)}{\sum_x p(x)q(t|x)} \right) \leq \sum_x p(x|y)q(t|x) \log \left(\frac{p(x|y)q(t|x)}{p(x)q(t|x)} \right). \quad (22)$$

So that, summing over y and t , we get $I_q(Y; T) \leq I(X; Y)$, with equality if and only if for all $y \in \text{supp}(p(Y)), t \in \text{supp}(q(T))$, it holds in (22). From the equality case of the log-sum inequality (Csiszár and Körner, 2011), the latter is equivalent to the existence of nonzero constants $(\alpha_{y,t})_{y \in \text{supp}(p(Y)), t \in \text{supp}(q(T))}$ such that

$$\forall x \in \mathcal{X}, \quad p(x)q(t|x) = \alpha_{y,t}p(x|y)q(t|x),$$

i.e., such that, for all $y \in \text{supp}(p(Y)), t \in \text{supp}(q(T))$, the quantity $\frac{p(x|y)}{p(x)}$ is constant on the subset of elements $x \in \mathcal{X}$ for which $q(t|x) > 0$. But the latter subset is precisely $\mathcal{X}_t^q$ (see definition (12)), and

$$\frac{p(x|y)}{p(x)} = \frac{1}{p(y)}p(y|x),$$

where $\frac{1}{p(y)}$ does not depend on x . Thus we proved that $I(Y; T) = I(X; Y)$ holds if and only if for all $t \in \text{supp}(q(T))$, the distribution $p(Y|x)$ does not depend on $x \in \mathcal{X}_t^q$: i.e., if and only if for all $t \in \text{supp}(q(T))$, there exists an $\mathcal{X}_j$ such that $\mathcal{X}_t^q \subseteq \mathcal{X}_j$.

(ii) $\Leftrightarrow$ (iii). Apply Lemma 7. ■

Combining the previous results directly yields that

$$\text{IB}(\Lambda) \subseteq E := \{\gamma \circ \pi_{\mathcal{X}}, \quad \gamma \in C_{\text{cong}}(\{1, \dots, n\}, \mathcal{T})\}.$$

Indeed, fix a solution $q(T|X) \in \text{IB}(\Lambda)$. Proposition 9 proves that $q(T|X) = \bar{q}(T|\mathcal{X}_J) \circ \pi_{\mathcal{X}}$. But because we must have $I_q(Y; T) = \Lambda := I(X; Y)$, Lemma 11 yields that $\bar{q}(T|\mathcal{X}_J)$ is here congruent.

Let us now prove the converse inclusion, i.e., that $E \subseteq \text{IB}(\Lambda)$.

Lemma 12 *For all $q(T|X) \in E$, we have $I_q(T; Y) = I(X; Y)$ and $I_q(X; T) = H(\pi_{\mathcal{X}}(X))$.*

Proof From point (i) in Lemma 8, $q(T|X) = \gamma \circ \pi_{\mathcal{X}}$ implies $\gamma = \bar{q}(T|\mathcal{X}_J)$. But by definition of E , here γ is assumed congruent, thus $\bar{q}(T|\mathcal{X}_J)$ is congruent. So that from Lemma 11, $I_q(T; Y) = I(X; Y)$. Point (ii) in Lemma 8 yields $I_q(X; T) = H(\pi_{\mathcal{X}}(X))$. ■

Now, because the IB problem is defined as the minimisation of a continuous function on a compact domain, it has at least one solution, say $q_*(T|X)$, which we know belongs to E from the inclusion $\text{IB}(\Lambda) \subseteq E$ that we already proved. But Lemma 12 then implies that for all $q(T|X) \in E$, we have $I_q(T; Y) = I_{q_*}(T; Y)$ and $I_q(X; T) = I_{q_*}(X; T)$. Thus any $q(T|X) \in E$ must also be a solution, i.e., $q(T|X) \in \text{IB}(\Lambda)$. This ends the proof of point (i) in Theorem 2.

B.3. End of the proof of Theorem 2

Proposition 9 ensures that for all λ and $q(T|X) \in \text{IB}(\lambda)$, we have the factorisation $q(T|X) = \bar{q}(T|\mathcal{X}_J) \circ \pi_{\mathcal{X}}$, with $\bar{q}(T|\mathcal{X}_J)$ defined in (9). Thus, for all $\sigma \in \text{Bij}(\mathcal{X})$,

$$\begin{aligned} \sigma \in G_{\text{ci}} &\Leftrightarrow \forall x \in \mathcal{X}, \quad p(Y|x) = p(Y|\sigma \cdot x) \\ &\Leftrightarrow \forall x \in \mathcal{X}, \quad x \sim_{\mathcal{X}} \sigma \cdot x \\ &\Leftrightarrow \pi \circ \sigma = \pi \\ &\Rightarrow \bar{q}(T|\mathcal{X}_J) \circ \pi \circ \sigma = \bar{q}(T|\mathcal{X}_J) \circ \pi \\ &\Leftrightarrow q(T|X) \circ \sigma = q(T|X), \end{aligned} \tag{23}$$

which yields point (iii) of Theorem 2. Moreover, if we assume that $\lambda = \Lambda$, then from Lemma 11, here $\bar{q}(T|\mathcal{X}_J)$ is a congruent channel, i.e., there exists a function f such that $f \circ \bar{q}(T|\mathcal{X}_J)$ is the identity on $\mathcal{X}$. Therefore the only implication in (23) becomes an equivalence as well, which yields point (ii) of Theorem 2.

Let us now prove point (iv) in Theorem 2. The statement is equivalent to proving that the equivalence relation defined by the partition in orbits under G_{ci} , which we denote here by $\sim_{\text{ci}}$, coincides with the equivalence relation $\sim_{\mathcal{X}}$ defined in (2). Moreover, by definition of an orbit, $x \sim_{\text{ci}} x'$ means that there exists $\sigma \in \text{Bij}(\mathcal{X})$ such that (i) $\sigma \in G_{\text{ci}}$, i.e., $p(Y|\sigma \cdot x'') = p(Y|x'')$ for all $x'' \in \mathcal{X}$, and (ii) $x' = \sigma \cdot x$.

Thus $x \sim_{\text{ci}} x'$ clearly implies $p(Y|x) = p(Y|x')$, i.e., $x \sim_{\mathcal{X}} x'$. Conversely, let us fix $x, x' \in \mathcal{X}$ such that $x \sim_{\mathcal{X}} x'$. We define σ as the transposition that permutes x and x' , and fixes all the other elements of $\mathcal{X}$. It is straightforward to verify that σ satisfies points (i) and (ii) above, i.e., that we have $x \sim_{\text{ci}} x'$.

Appendix C. Appendix for Section 3

C.1. On the projection on the exponential family

Let us recall that $\text{cl } \mathcal{E}$ denotes the topological closure of the exponential family $\mathcal{E}$. Here, we denote by $\tilde{p} \in \text{cl } \mathcal{E}$ the unique distribution (Ay et al., 2017) which achieves the minimum in $\inf_{r \in \text{cl } \mathcal{E}} D(p||r)$; but we do not assume, a priori, that $\tilde{p}$ minimises $\inf_{r \in \text{cl } \mathcal{E}} D(\kappa \cdot p || \kappa \cdot r)$. Note that we always have $\text{supp}(p) \subseteq \text{supp}(\tilde{p})$, because otherwise $D(p||\tilde{p}) = +\infty > \inf_{r \in \mathcal{E}} D(p||r)$.

In particular, whenever $p(A)$ is full support, then $\tilde{p}(A)$ is full support as well. In the latter case, $\tilde{p}$ is thus both in $\text{cl } \mathcal{E}$ and the interior of the simplex $\Delta_{\mathcal{A}}$, which implies that $\tilde{p} \in \mathcal{E}$.

Let us now prove that $\tilde{p}$ also achieves the latent space divergence, i.e., for all $\kappa \in C(\mathcal{A}, \mathcal{T})$, we automatically have $D(p||\tilde{p}) = \inf_{r \in \text{cl } \mathcal{E}} D(\kappa \cdot p||\kappa \cdot r)$. Indeed, for all $r \in \text{cl } \mathcal{E}$ and with the convention $0 \log(\frac{0}{0}) := 0$,

$$\begin{aligned}
 D(\kappa \cdot p||\kappa \cdot \tilde{p}) - D(\kappa \cdot p||\kappa \cdot r) &= \sum_{t \in \mathcal{T}} (\kappa \cdot p)(t) \log \left(\frac{(\kappa \cdot r)(t)}{(\kappa \cdot \tilde{p})(t)} \right) \\
 &= \sum_{t \in \mathcal{T}} \sum_{a \in \mathcal{A}} p(a) q(t|a) \log \left(\frac{\sum_{a \in \mathcal{A}} r(a) q(t|a)}{\sum_{a \in \mathcal{A}} \tilde{p}(a) q(t|a)} \right) \\
 &\leq \sum_{t \in \mathcal{T}} \sum_{a \in \mathcal{A}} p(a) q(t|a) \log \left(\frac{r(a) q(t|a)}{\tilde{p}(a) q(t|a)} \right) \quad (24) \\
 &= \sum_{a \in \mathcal{A}} p(a) \log \left(\frac{r(a)}{\tilde{p}(a)} \right) \\
 &= D(p||\tilde{p}) - D(p||r) \\
 &\leq 0, \quad (25)
 \end{aligned}$$

where line (24) uses the log-sum inequality (Csiszár and Körner, 2011), and line (25) that $D(p||\tilde{p}) = \inf_{r \in \text{cl } \mathcal{E}} D(p||r)$. In particular, $\tilde{p}$ is the unique distribution in $\text{cl } \mathcal{E}$ minimising both $\inf_{r \in \text{cl } \mathcal{E}} D(p||r)$ and $\inf_{r \in \text{cl } \mathcal{E}} D(\kappa \cdot p||\kappa \cdot r)$.

C.2. Proof of Theorem 3

We will use the notations and definitions from Section 3.1 and Appendix A, which are consistent. We also write $q(T, A)$, $q_{\pi}(A, T)$, $\tilde{q}(T, A)$, $\tilde{q}_{\pi}(T, A)$ the joint distributions defined resp. by $q(t, a) := p(a)q(t|a)$, $q_{\pi}(a, t) := p(a)q_{\pi}(t|a)$, $\tilde{q}(t, a) = \tilde{p}(a)q(t|a)$ and $\tilde{q}_{\pi}(t, a) = \tilde{p}(a)q_{\pi}(t|a)$. Note that for $\kappa = q(T|A)$, the quantity $D(\kappa \cdot p||\kappa \cdot \tilde{p})$ then becomes $D(q(T)||\tilde{q}(T))$, which will be a more convenient notation in the proofs below.

C.2.1. FACTORISATION FOR ALL PARAMETER λ

Proposition 13 *For all $0 \leq \lambda \leq \Lambda$, every solution $q(T|A) \in \text{DIB}(\lambda)$ factorises as $q(T|A) = \bar{q}(T|\mathcal{A}_J) \circ \pi$.*

The proof will derive from the following lemma:

Lemma 14 *For every $q(T|A) \in C(\mathcal{A}, \mathcal{T})$, we have:*

- (i) $q_{\pi}(T) = q(T)$ and $\tilde{q}_{\pi}(T) = \tilde{q}(T)$. In particular, $D(q_{\pi}(T)||\tilde{q}_{\pi}(T)) = D(q(T)||\tilde{q}(T))$.
- (ii) $I_{q_{\pi}}(A; T) \leq I_q(A; T)$, where equality holds if and only if $q_{\pi}(T|A) = q(T|A)$.

Proof (i). $q_\pi(T) = q(T)$ is point (i) in Lemma 6. Moreover, for all $t \in \mathcal{T}$,

$$\begin{aligned}
 \tilde{q}_\pi(t) &= \sum_{a \in \mathcal{A}} q_\pi(t|a) \tilde{p}(a) \\
 &= \sum_j \sum_{a \in \mathcal{A}_j} q_\pi(t|a) \tilde{p}(a) = \sum_j \sum_{a \in \mathcal{A}_j} \frac{q(t, \mathcal{A}_j)}{p(\mathcal{A}_j)} \tilde{p}(a) \\
 &= \sum_j \sum_{a, a' \in \mathcal{A}_j} \frac{1}{p(\mathcal{A}_j)} q(t|a') p(a') \tilde{p}(a) \\
 &= \sum_j \sum_{a, a' \in \mathcal{A}_j} \frac{1}{p(\mathcal{A}_j)} q(t|a') p(a) \tilde{p}(a') \\
 &= \sum_j \sum_{a' \in \mathcal{A}_j} q(t|a') \tilde{p}(a') = \sum_{a' \in \mathcal{A}} q(t|a') \tilde{p}(a') \\
 &= \tilde{q}(t)
 \end{aligned} \tag{26}$$

where (26) uses the definition of the sets $\mathcal{A}_j$ through the equivalence relation $\sim$ defined in (6), i.e., $p(a) \tilde{p}(a') = p(a) \tilde{p}(a')$ for $a, a' \in \mathcal{A}_j$.

(ii). We apply point (ii) in Lemma 6. ■

But Lemma 14 means that for the DIB problem (3), if we replace the channel $q(T|A)$ by the corresponding $q_\pi(T|A)$, then (i) the value for of the constraint function is unchanged, and (ii) the value of the target function does not increase, with equality if and only if $q_\pi(T|A) = q(T|A)$. In particular, if $q(T|A)$ solves the DIB problem, then we must have $q(T|A) = q_\pi(T|A)$: i.e., $q(T|A) = \bar{q}(T|\mathcal{A}_J) \circ \pi$.

C.2.2. EXPLICIT FORM OF SOLUTIONS FOR $\lambda = \Lambda$ (THEOREM 3)

In this section, we prove Theorem 3, i.e., that $\text{DIB}(\Lambda) = \{\gamma \circ \pi : \gamma \in C_{\text{cong}}(\bar{\mathcal{A}}, \mathcal{T})\}$. Recall that $\mathcal{A}_t^q$ is the “probabilistic pre-image of t through $q(T|A)$ ” (see equation (12)).

Lemma 15 *We have $D(q(T)||\tilde{q}(T)) \leq \Lambda := D(p(A)||\tilde{p}(A))$, and the following are equivalent:*

- (i) $D(q(T)||\tilde{q}(T)) = \Lambda$.
- (ii) For all $t \in \text{supp}(q(T))$, there exists a partition element $\mathcal{A}_j \in \bar{\mathcal{A}}$ such that $\mathcal{A}_t^q \subseteq \mathcal{A}_j$.
- (iii) The channel $\bar{q}(T|\mathcal{A}_J) \in C(\bar{\mathcal{A}}, \mathcal{T})$ defined in (9) is congruent.

Proof (i) $\Leftrightarrow$ (ii). We have

$$D(q(T)||\tilde{q}(T)) = \sum_{t \in \text{supp}(q(T))} \left(\sum_{a \in \mathcal{S}} q(t|a) p(a) \right) \log \left(\frac{\sum_{a \in \mathcal{S}} q(t|a) p(a)}{\sum_{a \in \mathcal{S}} q(t|a) \tilde{p}(a)} \right),$$

But for all $t \in \text{supp}(q(T))$, from the log-sum inequality (Csiszár and Körner, 2011), with the convention $0 \log(\frac{0}{0}) := 0$,

$$\left(\sum_{a \in \mathcal{S}} q(t|a)p(a) \right) \log \left(\frac{\sum_{a \in \mathcal{S}} q(t|a)p(a)}{\sum_{a \in \mathcal{S}} q(t|a)\tilde{p}(a)} \right) \leq \sum_{a \in \mathcal{S}} q(t|a)p(a) \log \left(\frac{q(t|a)p(a)}{q(t|a)\tilde{p}(a)} \right). \quad (27)$$

So that, summing over t , we get $D(q(T)||\tilde{q}(T)) \leq D(p(A)||\tilde{p}(A))$, with equality if and only if for all $t \in \mathcal{T}$, it holds in (27). From the equality case of the log-sum inequality (Csiszár and Körner, 2011), the latter is equivalent to the existence of nonzero constants $(\alpha_t)_{t \in \text{supp}(q(T))}$ such that

$$\forall a \in \mathcal{A} \quad q(t|a)p(a) = \alpha_t q(t|a)\tilde{p}(a),$$

i.e., such that, for all $t \in \text{supp}(q(T))$, we have $p(a) = \alpha_t \tilde{p}(a)$ for all $a \in \mathcal{A}$ such that $q(t|a) > 0$, i.e.,

$$\forall t \in \text{supp}(q(T)), \forall a \in \mathcal{A}_t^q, \frac{p(a)}{\tilde{p}(a)} = \alpha_t$$

In the above, note that the fraction $\frac{p(a)}{\tilde{p}(a)}$ does make sense, because here $\text{supp}(\tilde{p}) = \mathcal{A}$ (see Section C.1). Thus we proved that equality holds in (27) if and only if for all $t \in \text{supp}(q(T))$, the quotient $p(a)/\tilde{p}(a)$ is constant on $\mathcal{A}_t^q$, i.e., if and only if for all $t \in \text{supp}(q(T))$, there exists an $\mathcal{A}_j$ such that $\mathcal{A}_t^q \subseteq \mathcal{A}_j$.

(ii) $\Leftrightarrow$ (iii). Apply Lemma 7. ■

Combining the previous results directly yields that

$$\text{DIB}(\Lambda) \subseteq E := \{ \gamma \circ \pi, \quad \gamma \in C_{\text{cong}}(\bar{\mathcal{A}}, \mathcal{T}) \}.$$

Indeed, fix a solution $q(T|A) \in \text{DIB}(\Lambda)$. Proposition 13 proves that $q(T|A) = \bar{q}(T|\mathcal{A}_J) \circ \pi$. But because we must have $D(q(T)||\tilde{q}(T)) = \Lambda := D(p(A)||\tilde{p}(A))$, Lemma 15 yields that $\bar{q}(T|\mathcal{A}_J)$ is here congruent.

Let us now prove the converse inclusion, i.e., that $E \subseteq \text{DIB}(\Lambda)$.

Lemma 16 *For all $q(T|A) \in E$, we have $D(q(T)||\tilde{q}(T)) = D(p(A)||\tilde{p}(A))$ and $I_q(A; T) = H(\pi(A))$.*

Proof From point (i) in Lemma 8, $q(T|A) = \gamma \circ \pi$ implies $\gamma = \bar{q}(T|\mathcal{A}_J)$. But by definition of E , here γ is assumed congruent, thus so is $\bar{q}(T|\mathcal{A}_J)$. So that from Lemma 15, $D(q(T)||\tilde{q}(T)) = \Lambda = D(p(A)||\tilde{p}(A))$. Point (ii) in Lemma 8 yields $I_q(A; T) = H(\pi(A))$. ■

Now, because the DIB problem is defined as the minimisation of a continuous function on a compact domain, it has at least one solution, say $q_*(T|A)$, which we know belongs to E from the inclusion $\text{DIB}(\Lambda) \subseteq E$ that we already proved. But Lemma 16 then implies that for all $q(T|A) \in E$, we have $D(q(T)||\tilde{q}(T)) = D(q_*(T)||\tilde{q}_*(T))$ and $I_q(A; T) = I_{q_*}(A; T)$. Thus any $q(T|A) \in E$ must also be a solution, i.e., $q(T|A) \in \text{DIB}(\Lambda)$. This ends the proof of point (i) in Theorem 3.

C.3. The Divergence IB captures equivariances (proof of Theorem 4)

(i). The proof is almost identical to that of point (ii) of Theorem 2 (see Appendix B.3). Here $\text{DIB}(\Lambda)$ denotes the solutions to the specific DIB problem defined in Section 3.2, and π the corresponding projection defined by

$$(x, y) \sim (x', y') \Leftrightarrow p(x, y)p(x') = p(x', y')p(x).$$

Proposition 13 ensures that for all λ and $\kappa = q(T|A) \in \text{DIB}_{\text{ce}}(\lambda)$, we have the factorisation $\kappa = \bar{\kappa} \circ \pi$, with $\bar{\kappa} = \bar{q}(T|\mathcal{S}_J)$ defined in (9). Thus, for all $(\sigma, \tau) \in \text{Bij}(\mathcal{X}) \times \text{Bij}(\mathcal{Y})$,

$$\begin{aligned} (\sigma, \tau) \in G_{\text{ce}} &\Leftrightarrow \forall (x, y) \in \mathcal{X} \times \mathcal{Y}, \quad p(y|x) = p(\tau \cdot y|\sigma \cdot x) \\ &\Leftrightarrow \forall (x, y) \in \mathcal{X} \times \mathcal{Y}, \quad p(x, y)p(\sigma \cdot x) = p(\sigma \cdot x, \tau \cdot y)p(x) \\ &\Leftrightarrow \forall (x, y) \in \mathcal{X} \times \mathcal{Y}, \quad (x, y) \sim (\sigma \cdot x, \tau \cdot y) \\ &\Leftrightarrow \pi \circ (\sigma, \tau) = \pi \\ &\Rightarrow \bar{\kappa} \circ \pi \circ (\sigma, \tau) = \bar{\kappa} \circ \pi \\ &\Leftrightarrow \kappa \circ (\sigma, \tau) = \kappa, \end{aligned} \tag{28}$$

where the first equivalence follows easily from the definition of equivariance (see Lemma 15 in (Charvin et al., 2023) for details). This yields point (ii). Moreover, if we assume that $\lambda = \Lambda$, then from Lemma 15, here $\bar{\kappa}$ is a congruent channel, i.e., there exists a function f such that $f \circ \bar{\kappa}$ is the identity on $\mathcal{A}$. Thus the only implication in (28) becomes an equivalence as well, which yields point (i) of Theorem 4.

(iii). Here, the reasoning used for the proof of point (iii) in Theorem 2 does not work. Indeed the transposition $\Phi \in \text{Bij}(\mathcal{X} \times \mathcal{Y})$ that permutes two pairs (x, y) and (x', y') and fixes all the other ones does not have a split form $\sigma \otimes \tau$ for some $(\sigma, \tau) \in \text{Bij}(\mathcal{X}) \times \text{Bij}(\mathcal{Y})$.

Moreover, let $\mathcal{X} = \{1, 2, 3\}$ and $\mathcal{Y} = \{1, 2\}$, with $p(X)$ uniform and $p(Y|X)$ defined through the row transition matrix

$$\begin{pmatrix} c & p_{12} \\ p_{21} & c \\ p_{31} & p_{32} \end{pmatrix}$$

where we choose $c, p_{12}, p_{21}, p_{31}$, and p_{32} pairwise distinct. It can be easily shown that this channel has no non-trivial equivariances, i.e., $G_{\text{ce}} = \{e_{\mathcal{X} \times \mathcal{Y}}\}$, so that the projection on orbits π_{ce} is the identity of $\mathcal{X} \times \mathcal{Y}$. Yet the projection π defined by the relation $\sim$ will here identify the two pairs (x, y) such that $p(y|x) = c$. Therefore $\pi_{\text{ce}} \neq \pi$.

C.4. Relation to the Intertwining IB

Our work is heavily inspired from that in (Charvin et al., 2023); in this section we explicitly relate the two. The latter reference considered the Intertwining IB problem, namely,

$$\text{IIB}(\lambda) := \arg \min_{\substack{\kappa \in C(\mathcal{X} \times \mathcal{Y}, \mathcal{T}) : \\ D(\kappa \cdot p(X, Y) || \kappa \cdot p(X)p(Y)) \geq \lambda}} I_{\kappa}(X, Y; T). \tag{29}$$

It can readily be verified this is a DIB problem with $\mathcal{E} = \Delta_{\mathcal{X}} \otimes \Delta_{\mathcal{Y}}$ and $C = C(\mathcal{X} \times \mathcal{Y}, \mathcal{T})$.

Problem (29) is used in (Charvin et al., 2023) to characterise equivariances under specific conditions: if (i) the distribution $p(X, Y)$ is discrete and full support, and (ii) $p(Y)$ is uniform, then the solution κ to (30) with $\lambda = I(X; Y)$ are such that a pair $(\sigma, \tau) \in \text{Bij}(\mathcal{X}) \times \text{Bij}(\mathcal{Y})$ is an equivariance if and only if $\kappa \circ (\sigma \otimes \tau) = \kappa$.

Thus, Section 3.2 in this work is an improvement on the latter result: here, we replace the Intertwining IB problem by the similar problem DIB_{ce} , from which we obtain the same characterisation as above, except that the assumption (ii) that $p(Y) = \mathcal{U}(\mathcal{Y})$ can now be dropped. Note that the latter assumption holds precisely when the hierarchical model $\mathcal{E}_{\text{IB}} := \Delta_{\mathcal{X}} \otimes \Delta_{\mathcal{Y}}$ from (Charvin et al., 2023) coincides with the hierarchical model $\mathcal{E}_{\text{ce}} := \Delta_{\mathcal{X}} \otimes \{\mathcal{U}(\mathcal{Y})\}$ from the present article. In this sense, we have clarified that $\mathcal{E}_{\text{ce}}$, and not $\mathcal{E}_{\text{IB}}$, is the “correct” hierarchical model to characterise — and soften — equivariances.

More broadly, the present work generalises the Intertwining IB framework to arbitrary (discrete) underlying space $\mathcal{A}$ and arbitrary hierarchical model $\mathcal{E}$. Note that our definition of soft equivariances is directly analogous to that proposed in (Charvin et al., 2023), even though it is different, as it based on a DIB problem defined by $\mathcal{E}_{\text{ce}}$ and not $\mathcal{E}_{\text{IB}}$.

Eventually, whereas (Charvin et al., 2023) does not provide algorithms to compute solutions to the IIB problem (29), here we design a BA algorithm for a general DIB problem (without constraints on encoder shape) with the same convergence guarantees as the BA for the classic IB (see Section D.2), which we use for synthetic numerical experiments in Section 4.

C.5. The classic IB is a Divergence IB

Ref. (Charvin et al., 2023) proves that the classic IB can be formulated as an Intertwining IB with specific constraints on the shape of compression channels. More precisely, define $\mathcal{T}$ as $\mathcal{T} := \mathcal{T}_{\text{IB}} \times \mathcal{Y}$ with $\mathcal{T}_{\text{IB}} := \mathbb{N}$,⁶ and consider the set

$$C_{\text{IB}(X, Y)} := \{\kappa_{\mathcal{X}} \otimes e_{\mathcal{Y}} : \kappa_{\mathcal{X}} \in C(\mathcal{X}, \mathcal{T}_{\text{IB}})\} \subset C(\mathcal{X} \times \mathcal{Y}, \mathcal{T}_{\text{IB}} \times \mathcal{Y})$$

of channels that can compress the $\mathcal{X}$ coordinate but copy the $\mathcal{Y}$ coordinate. This leads to the problem

$$\text{IIB}_{C_{\text{IB}}}(\lambda) := \arg \min_{\substack{\kappa \in C_{\text{IB}(X, Y)} : \\ D(\kappa(p(X, Y)) \| \kappa(p(X)p(Y))) = \lambda}} I_{\kappa}(X, Y; T), \quad (30)$$

Then:

Proposition 17 ((Charvin et al., 2023), Prop. 5) *For every $0 \leq \lambda \leq I(X; Y)$, a channel $\kappa_{\mathcal{X}} \otimes e_{\mathcal{Y}} \in C_{\text{IB}(X, Y)}$ solves the problem (30) if and only if $\kappa_{\mathcal{X}} = \kappa_{\mathcal{X}}(T_{\text{IB}}|X)$ solves the IB problem (1).*

In this sense, the classic IB is equivalent to the problem (30). Importantly, it can be easily verified that the latter is a DIB with $C = C_{\text{IB}(X, Y)}$ and $\mathcal{E} = \Delta_{\mathcal{X}} \otimes \Delta_{\mathcal{Y}}$.

However, for the sake of consistency with the results presented in this work, let us also prove that the classic IB is equivalent to a DIB with still $C = C_{\text{IB}(X, Y)}$, but now

$$\mathcal{E} = \mathcal{E}_{\text{ce}} := \{r(X)\mathcal{U}(\mathcal{Y}), r(X) \in \Delta_{\mathcal{X}}\},$$

6. This choice is formally equivalent to $\mathcal{T} := \mathbb{N}$, as there is a bijection between $\mathbb{N}$ and $\mathbb{N} \times \mathcal{Y}$.

which is the exponential family used in Section 3.2 to fully characterise channel equivariances.

As mentioned in Section 3.2, we have $D(p(X, Y) || \mathcal{E}_{ce}) = D(p(X, Y) || p(X)\mathcal{U}(\mathcal{Y}))$. Moreover for $\kappa = \kappa_X \otimes e_Y \in C_{IB}$, we have $\kappa \cdot p(X, Y) = q(T, Y)$, while $\kappa \cdot p(X)\mathcal{U}(\mathcal{Y}) = q(T)\mathcal{U}(\mathcal{Y})$ and $\kappa \cdot p(X)p(Y) = q(T)p(Y)$, where the joint distribution $q(T, X, Y)$ is defined using $\kappa = q(T|X)$ together with $p(X, Y)$ and the Markov chain $T - X - Y$. Thus

$$\begin{aligned} D(\kappa \cdot p(X, Y) || \kappa \cdot \mathcal{E}_{ce}) &= D(q(T, Y) || q(T)\mathcal{U}(\mathcal{Y})) \\ &= D(q(T, Y) || q(T)p(Y)) + D(p(Y) || \mathcal{U}(\mathcal{Y})) \\ &= D(\kappa(p(X, Y)) || \kappa(p(X)p(Y))) + D(p(Y) || \mathcal{U}(\mathcal{Y})). \end{aligned}$$

Therefore the constraint function in the DIB defined in (30), and the constraint function for the same problem but with $\mathcal{E}_{IB}$ replaced by $\mathcal{E}_{ce}$, differ by a constant K that depends only on $p(Y)$, which is here fixed. In particular, the corresponding DIB problems are equivalent, in that for all $0 \leq \lambda \leq D(p(X, Y) || p(X)p(Y))$,

$$\text{IIB}_{C_{IB}}(\lambda) = \text{DIB}_{\mathcal{E}_{ce}, C_{IB}}(\lambda + K).$$

As Proposition 17 proves that $\text{IIB}_{C_{IB}}$ is equivalent to the classic IB, this proves that $\text{DIB}_{\mathcal{E}_{ce}, C_{IB}}$ is also equivalent to the classic IB (up to shifting the trade-off parameter λ by a constant K).

In particular, our framework captures channel invariances — which are a special case of channel equivariances with trivial action on the output space — by using the exponential family $\mathcal{E}_{ce}$ that captures equivariances, and imposing the additional constraint C_{IB} of only compressing the input space but leaving the output space unchanged.

C.6. The Divergence IB captures distribution invariances (proof of Theorem 5)

The proof is almost identical to that of points (ii) and (iii) of Theorem 2 (see Appendix B.3). Here $\text{DIB}(\Lambda)$ denotes the solutions to the specific DIB problem defined in Section 3.3, and π the corresponding projection defined by

$$a \sim a' \Leftrightarrow p(a) = p(a') \quad (31)$$

Proposition 13 ensures that for all λ and $\kappa \in \text{DIB}(\lambda)$, we have the factorisation $\kappa = \bar{\kappa} \circ \pi$, with $\bar{\kappa}$ defined in (9). Thus, for all $\Phi \in \text{Bij}(\mathcal{A})$,

$$\begin{aligned} \Phi \in G_{\text{di}} &\Leftrightarrow \forall a \in \mathcal{A}, \quad p(a) = p(\Phi \cdot a) \\ &\Leftrightarrow \forall a \in \mathcal{A}, \quad a \sim \Phi \cdot a \\ &\Leftrightarrow \pi \circ \Phi = \pi \\ &\Rightarrow \bar{\kappa} \circ \pi \circ \Phi = \bar{\kappa} \circ \pi \\ &\Leftrightarrow \kappa \circ \Phi = \kappa. \end{aligned} \quad (32)$$

This yields point (iii) of Theorem 3. Moreover, if we assume that $\lambda = \Lambda$, then from Lemma 15, here $\bar{\kappa}$ is a congruent channel, i.e., there exists a function f such that $f \circ \bar{\kappa}$ is the identity on $\bar{\mathcal{S}}$. Therefore the only implication in (32) becomes an equivalence as well, which yields point (i).

(iii). The statement is equivalent to proving that the equivalence relation defined by the partition in orbits under G_{di} , which we denote here by $\sim_{\text{di}}$, coincides with the equivalence relation $\sim$ defined in (31). Moreover, by definition of an orbit, $a \sim_{\text{di}} a'$ means that there exist $\Phi \in G_{\text{di}}$ such that (i) $\Phi \in G_{\text{di}}$, i.e., $p(\Phi \cdot a'') = p(a'')$ for all $a'' \in \mathcal{A}$, and (ii) $a' = \Phi \cdot a$.

Thus $a \sim_{\text{di}} a'$ clearly implies $p(a) = p(a')$, i.e., $a \sim a'$. Conversely, let us fix $a, a' \in \mathcal{A}$ such that $a \sim a'$. We define $\Phi \in \text{Bij}(\mathcal{A})$ as the transposition that permutes a and a' , and fixes all the other elements of $\mathcal{A}$. It is straightforward to verify that Φ satisfies points (i) and (ii) above, i.e., that we have $a \sim_{\text{di}} a'$.

Appendix D. Appendix for section 3.4

In this appendix, the distribution $p(A)$ is allowed to not be full support, and we denote by $\mathcal{S}$ this support. In this case, there is still a unique distribution $\tilde{p}$ in the closure $\text{cl } \mathcal{E}$ of $\mathcal{E}$ such that $D(p||\tilde{p}) = \inf_{r \in \mathcal{E}} D(p||r)$ (Ay et al., 2017). We denote by $\tilde{\mathcal{S}}$ the support of $\tilde{p}$. Note that $D(p||\tilde{p}) = \inf_{r \in \mathcal{E}} D(p||r) < +\infty$ implies $\mathcal{S} \subseteq \tilde{\mathcal{S}}$. We also assume now that $\mathcal{T}$ is finite, and we define the DIB Lagrangian, on $C(\mathcal{A}, \mathcal{T})$, as

$$\mathcal{L}_\beta(q(T|A)) := I_q(A; T) - \beta D(q(T)||\tilde{q}(T)). \quad (33)$$

After technical preliminaries in Section D.1, we prove in Section D.2.1 the following necessary condition for local minimisers $\kappa = q(T|A)$ of (33): for all $a \in \text{supp}(p(A))$ and $t \in \text{supp}(q(T))$,

$$q(t|a) = \frac{1}{Z(a, \beta)} q(t) \exp \left[-\beta \left(\frac{q(t)\tilde{p}(a)}{\tilde{q}(t)p(a)} - \log \left(\frac{q(t)\tilde{p}(a)}{\tilde{q}(t)p(a)} \right) - 1 \right) \right], \quad (34)$$

where $q(t) := \sum_a p(a)q(t|a)$ and $\tilde{q}(t) := \sum_a \tilde{p}(a)q(t|a)$, with $Z(a, \beta)$ a normaliser. From this fixed-point equation, we obtain a Blahut-Arimoto (BA) algorithm with the same guarantees as BA for the classic IB (Tishby et al., 2000) (see Section D.2.2). Eventually, we provide more details on effective cardinality in Section D.3.

D.1. Minimisers on $\mathcal{S} \subseteq \mathcal{A}$ yield minimisers on $\mathcal{A}$

In this section, we reduce the minimisation of $\mathcal{L}_\beta$ on $C(\mathcal{A}, \mathcal{T})$ to a minimisation over channels defined only on the support $\mathcal{S}$ of $p = p(A)$. More precisely, we show that a minimiser of $\mathcal{L}_\beta$ can always be obtained the following way: choose a minimiser $\kappa \in C(\mathcal{S}, \mathcal{T})$ of the Lagrangian $\mathcal{L}_\beta$ restricted to $C(\mathcal{S}, \mathcal{T})$, and extend it to a channel in $C(\mathcal{A}, \mathcal{T})$ by sending $\mathcal{A} \setminus \mathcal{S}$ on a dummy symbol $t_0 \notin \text{supp}(\kappa \cdot p)$. This allows us, in our numerical experiments, to reduce the computation of minimisers of (33) to that of the same Lagrangian restricted to channels defined only on the support $\mathcal{S} \subseteq \mathcal{A}$. This restriction to the support allows us to then use the BA algorithm (see Section D.2), which indeed can only be applied on the support $\mathcal{S}$.

Let $q(T|A) \in C(\mathcal{A}, \mathcal{T})$. We write $q_{\mathcal{S}}(T|A) \in C(\mathcal{S}, \mathcal{T})$ and $p_{\mathcal{S}}(A)$ the restrictions of $q(T|A)$, resp. $p(A)$, to $\mathcal{S} \subseteq \mathcal{A}$: i.e., $q_{\mathcal{S}}(t|a) := q(t|a)$ and $p_{\mathcal{S}}(a) := p(a)$ for all $a \in \mathcal{S}$, $t \in \mathcal{T}$ — note that these are abuses of notation, as the input alphabet of $q_{\mathcal{S}}(T|A)$ is actually only $\mathcal{S}$, and similarly $p_{\mathcal{S}}(A)$ is only defined on $\mathcal{S}$. Of course $p_{\mathcal{S}}$ is a probability on $\mathcal{S} := \text{supp}(p(A))$. We extend all the notations relating to $q(T|A)$ in Section C to $q_{\mathcal{S}}(T|A)$; in particular, for

$a \in \mathcal{S}, t \in \mathcal{T}$,

$$\begin{aligned} q_{\mathcal{S}}(t) &:= \sum_{a \in \mathcal{S}} q_{\mathcal{S}}(t, a) := \sum_{a \in \mathcal{S}} p_{\mathcal{S}}(a) q_{\mathcal{S}}(t|a) = \sum_{a \in \mathcal{S}} p(a) q(t|a) = q(t), \\ \tilde{q}_{\mathcal{S}}(t) &:= \sum_{a \in \mathcal{S}} \tilde{q}_{\mathcal{S}}(t, a) := \sum_{a \in \mathcal{S}} \tilde{p}_{\mathcal{S}}(a) q_{\mathcal{S}}(t|a) := \sum_{a \in \mathcal{S}} \tilde{p}(a) q(t|a), \end{aligned} \quad (35)$$

or

$$\begin{aligned} \mathcal{L}_{\beta, \mathcal{S}}(q_{\mathcal{S}}(T|A)) &:= I_{q_{\mathcal{S}}}(A; T) - \beta D(q_{\mathcal{S}}(T) || \tilde{q}_{\mathcal{S}}(T)) \\ &:= \sum_{a \in \mathcal{S}, t \in \text{supp}(q_{\mathcal{S}}(T))} p_{\mathcal{S}}(a) q_{\mathcal{S}}(t|a) \log \left(\frac{q_{\mathcal{S}}(t|a)}{q_{\mathcal{S}}(t)} \right) - \beta \sum_{t \in \text{supp}(q_{\mathcal{S}}(T))} q_{\mathcal{S}}(t) \log \left(\frac{q_{\mathcal{S}}(t)}{\tilde{q}_{\mathcal{S}}(t)} \right). \end{aligned} \quad (36)$$

We also denote by $\tilde{\mathcal{S}} \subseteq \mathcal{A}$ the support of $\tilde{p} = \tilde{p}(A)$.

Proposition 18 *Let $q(T|A) \in C(\mathcal{A}, \mathcal{T})$. Then $q(T|A)$ is a global minimum of $\mathcal{L}_{\beta}$ if and only if*

(i) $q_{\mathcal{S}}(T|A)$ is a global minimum of $\mathcal{L}_{\beta, \mathcal{S}}$,

(ii) For all $t \in \text{supp}(q(T))$ and $a \in \tilde{\mathcal{S}} \setminus \mathcal{S}$, we have $q(t|a) = 0$.

In particular, if $q_{\mathcal{S}}(T|A)$ is a global minimum $\mathcal{L}_{\beta, \mathcal{S}}$, we obtain a global minimum of $\mathcal{L}_{\beta}$ with the extension $q'(T|A) \in C(\mathcal{A}, \mathcal{T})$ of $q_{\mathcal{S}}(T|A)$ defined through

$$q'(T|a) := \begin{cases} q_{\mathcal{S}}(T|a) & \text{if } a \in \mathcal{S}, \\ \delta_{t_0} & \text{if } a \in \mathcal{A} \setminus \mathcal{S}, \end{cases} \quad (37)$$

where we chose $t_0 \in \mathcal{T} \setminus \text{supp}(q(T))$.

Before proving this result, let us recall that $q(t) = \sum_{a \in \mathcal{S}} p(a) q(t|a)$, so that $\text{supp}(q(T)) = \text{supp}(q_{\mathcal{S}}(T))$ can be seen as the “probabilistic image of $\mathcal{S}$ through the channel $q_{\mathcal{S}}(T|A)$ ”, and does not depend on the values of $q(t|a)$ for $a \in \tilde{\mathcal{S}} \setminus \mathcal{S}$. Thus the condition (ii) in Proposition 18 means that $q(T|A)$ sends the elements of $\mathcal{S}$ and $\tilde{\mathcal{S}} \setminus \mathcal{S}$ on distinct subsets of bottleneck symbols in $\mathcal{T}$. Moreover, intuitively, the channel $q'(T|A)$ extends $q_{\mathcal{S}}(T|A)$ by sending all the elements a outside $\mathcal{S}$ on a “dummy” symbol t_0 which lies outside the image $\text{supp}(q_{\mathcal{S}}(T))$ of $\mathcal{S}$ through $q_{\mathcal{S}}(T|A)$.

Proof We have

$$\begin{aligned}
 D(q(T)||\tilde{q}(T)) &= \sum_{t \in \text{supp}(q(T))} q(t) \log \left(\frac{q(t)}{\tilde{q}(t)} \right) \\
 &= \sum_{t \in \text{supp}(q(T)), a \in \mathcal{A}} q(t|a)p(a) \log \left(\frac{\sum_{a \in \mathcal{A}} q(t|a)p(a)}{\sum_{a \in \mathcal{A}} q(t|a)\tilde{p}(a)} \right) \\
 &= \sum_{t \in \text{supp}(q(T)), a \in \mathcal{S}} q(t|a)p(a) \log \left(\frac{\sum_{a \in \mathcal{S}} q(t|a)p(a)}{\sum_{a \in \mathcal{S}} q(t|a)\tilde{p}(a) + \sum_{a \in \tilde{\mathcal{S}} \setminus \mathcal{S}} q(t|a)\tilde{p}(a)} \right) \\
 &\leq \sum_{t \in \text{supp}(q(T)), a \in \mathcal{S}} q(t|a)p(a) \log \left(\frac{\sum_{a \in \mathcal{S}} q(t|a)p(a)}{\sum_{a \in \mathcal{S}} q(t|a)\tilde{p}(a)} \right) \tag{38}
 \end{aligned}$$

$$\begin{aligned}
 &= \sum_{t \in \text{supp}(q(T)), a \in \mathcal{S}} q'(t|a)p(a) \log \left(\frac{\sum_{a \in \mathcal{S}} q'(t|a)p(a)}{\sum_{a \in \mathcal{S}} q'(t|a)\tilde{p}(a)} \right) \\
 &= \sum_{t \in \text{supp}(q(T)), a \in \mathcal{A}} q'(t|a)p(a) \log \left(\frac{\sum_{a \in \mathcal{A}} q'(t|a)p(a)}{\sum_{a \in \mathcal{A}} q'(t|a)\tilde{p}(a)} \right) \tag{39}
 \end{aligned}$$

$$\begin{aligned}
 &= \sum_{t \in \mathcal{T}, a \in \mathcal{A}} q'(t|a)p(a) \log \left(\frac{\sum_{a \in \mathcal{A}} q'(t|a)p(a)}{\sum_{a \in \mathcal{A}} q'(t|a)\tilde{p}(a)} \right) \tag{40} \\
 &= D(q'(T)||\tilde{q}'(T)),
 \end{aligned}$$

where we defined with the marginals $q'(T) := \sum_{t \in \mathcal{T}} q'(t|a)p(a)$ and $\tilde{q}'(T) := \sum_{t \in \mathcal{T}} q'(t|a)\tilde{p}(a)$. Note that (39) uses $q'(t|a) = 0$ for $a \in \mathcal{A} \setminus \mathcal{S}$, $t \in \text{supp}(q(T))$, and (40) uses the definition (37) of $q'(T|A)$, which implies

$$q'(t_0|a)p(a) = (0 \times p(a)) \delta_{a \in \mathcal{S}} + (q'(t_0|a) \times 0) \delta_{a \in \mathcal{A} \setminus \mathcal{S}} = 0,$$

and thus

$$\sum_{a \in \mathcal{A}} q'(t_0|a)p(a) \log \left(\frac{\sum_{a \in \mathcal{A}} q'(t_0|a)p(a)}{\sum_{a \in \mathcal{A}} q'(t_0|a)\tilde{p}(a)} \right) = 0.$$

Moreover, the r.h.s. of (38) coincides with $D(q_{\mathcal{S}}(T)||q_{\mathcal{S}}(T))$. On the other hand it is straightforward to verify that $I_q(A; T) = I_{q'}(A; T) = I_{q_{\mathcal{S}}}(A; T)$. Thus

$$\mathcal{L}_{\beta}(q(T|A)) \geq \mathcal{L}_{\beta}(q'(T|A)) = \mathcal{L}_{\beta, \mathcal{S}}(q_{\mathcal{S}}(T|A)), \tag{41}$$

and equality is achieved in (41) if and only if it is achieved in (38). The latter is equivalent to $\sum_{a \in \tilde{\mathcal{S}} \setminus \mathcal{S}} q(t|a)\tilde{p}(a) = 0$ for all $t \in \text{supp}(q(T))$, i.e., to $q(t|a) = 0$ for all $t \in \text{supp}(q(T))$ and $a \in \tilde{\mathcal{S}} \setminus \mathcal{S}$, i.e., to point (ii) in Proposition 18.

Assume now that $q(T|A)$ minimises $\mathcal{L}_{\beta}$. Then equation (41) and its equality case clearly imply point (ii) in Proposition 18. Moreover, if $q_{\mathcal{S},1}(T|A)$ is another channel in $C(\mathcal{S}, \mathcal{T})$, we can extend it to a channel $q'_1(T|A)$ similarly as in (37), which yields

$$\mathcal{L}_{\beta, \mathcal{S}}(q_{\mathcal{S},1}(T|A)) = \mathcal{L}_{\beta}(q'_1(T|A)) \geq \mathcal{L}_{\beta}(q(T|A)) = \mathcal{L}_{\beta}(q'(T|A)) = \mathcal{L}_{\beta, \mathcal{S}}(q_{\mathcal{S}}(T|A)),$$

whence point (i) in Proposition 18.

Conversely, assume that points (i) and (ii) hold. Fix an arbitrary distribution $q_1(T|A) \in C(\mathcal{A}, \mathcal{T})$ and write $q_{\mathcal{S},1}(T|A) \in C(\mathcal{S}, \mathcal{T})$, resp. $q'_1(T|A) \in C(\mathcal{A}, \mathcal{T})$, the restriction of $q_1(T|A)$ to $\mathcal{S}$, resp. the corresponding channel defined similarly as in (37). Then

$$\begin{aligned} \mathcal{L}_\beta(q_1(T|A)) &\geq \mathcal{L}_\beta(q'_1(T|A)) = \mathcal{L}_{\beta,\mathcal{S}}(q_{\mathcal{S},1}(T|A)) \\ &\geq \mathcal{L}_{\beta,\mathcal{S}}(q_{\mathcal{S}}(T|A)) = \mathcal{L}_\beta(q'(T|A)) = \mathcal{L}_\beta(q(T|A)), \end{aligned}$$

where the last equality uses point (ii) and the equality case of (41). Therefore $q(T|A)$ is indeed a global minimum of $\mathcal{L}_\beta$. $\blacksquare$

D.2. Self-consistent equation and Blahut-Arimoto algorithm

Here we describe a Blahut-Arimoto (BA) iterative algorithm to compute the minimisers of the DIB Lagrangian (33). Following Proposition 18, we aim at a minimiser $q_{\mathcal{S}}(T|A)$ of the Lagrangian $\mathcal{L}_{\beta,\mathcal{S}}$ restricted to $\mathcal{S} := \text{supp}(p(A))$ (see equation (36)), which automatically yields solutions for channels defined on the whole alphabet $\mathcal{A}$ (see equation (37)). To alleviate notations, in this section we will directly write $q(T|A)$ and $\mathcal{L}_\beta$ instead of $q_{\mathcal{S}}(T|A)$ and $\mathcal{L}_{\beta,\mathcal{S}}$. As we will see, our algorithm does not provably converge to a global minimum of the DIB Lagrangian, but it has the same guarantees as the BA algorithm for the classic IB (Tishby et al., 2000): namely, the values of the Lagrangian decrease at each step and converge to a fixed value, and the limit of a corresponding convergent sequence $(\kappa_i)_{i \in \mathbb{N}}$ must satisfy equation (34).

D.2.1. CRITICAL POINTS ARE CHARACTERISED BY A SELF-CONSISTENT EQUATION

Taking into account the constraints $\sum_{t \in \mathcal{T}} q(t|a) = 1$ for all $a \in \mathcal{S}$, but not the inequality constraints $q(t|a) \geq 0$ for all $a \in \mathcal{S}, t \in \mathcal{T}$, we obtain the extended Lagrangian

$$\mathcal{L}_{\beta,\mu}(q(T|A)) := I_q(A; T) - \beta D(q(T) || \tilde{q}(T)) + \sum_{a \in \mathcal{S}, t \in \mathcal{T}} \mu_a q(t|a). \quad (42)$$

We derive $\mathcal{L}_{\beta,\mu}$ on the open set

$$\mathcal{Q}_+ := \{(q(t|a))_{a \in \mathcal{S}, t \in \mathcal{T}} : \forall a \in \mathcal{S}, \forall t \in \mathcal{T}, q(t|a) > 0\} = (\mathbb{R}_+)^{|\mathcal{S}||\mathcal{T}|}.$$

First, $q(t) := \sum_{a'} p(a') q(t|a')$ and $\tilde{q}(t) := \sum_{a'} \tilde{p}(a') q(t|a')$, so that

$$\begin{aligned} \partial_{q(t|a)} q(t) &= p(a), \\ \partial_{q(t|a)} \tilde{q}(t) &= \tilde{p}(a). \end{aligned}$$

Moreover, note that $q(T)$ and $\tilde{q}(T)$ are strictly positive for $(q(t|a))_{a,t} \in \mathcal{Q}_+$. Thus we can write

$$\begin{aligned}
 \partial_{q(t|a)} I_q(A; T) &= \partial_{q(t|a)} \sum_{a', t'} p(a') q(t'|a') \log \left(\frac{q(t|a')}{q(t)} \right) \\
 &= p(a) \log \left(\frac{q(t|a)}{q(t)} \right) + \sum_{a'} p(a') q(t|a') \frac{q(t)}{q(t|a')} \frac{q(t) \delta_{a'=a} - p(a) q(t|a')}{q(t)^2} \\
 &= p(a) \log \left(\frac{q(t|a)}{q(t)} \right) + \sum_{a'} p(a') \left(\delta_{a'=a} - \frac{p(a) q(t|a')}{q(t)} \right) \\
 &= p(a) \log \left(\frac{q(t|a)}{q(t)} \right) + p(a) - p(a) \frac{q(t)}{q(t)} \\
 &= p(a) \log \left(\frac{q(t|a)}{q(t)} \right),
 \end{aligned}$$

and

$$\begin{aligned}
 \partial_{q(t|a)} D(q(T) || \tilde{q}(T)) &= \partial_{q(t|a)} \sum_{a', t'} p(a') q(t'|a') \log \left(\frac{q(t)}{\tilde{q}(t)} \right) \\
 &= p(a) \log \left(\frac{q(t)}{\tilde{q}(t)} \right) + \sum_{a'} p(a') q(t|a') \frac{\tilde{q}(t)}{q(t)} \frac{p(a) \tilde{q}(t) - \tilde{p}(a) q(t)}{\tilde{q}(t)^2} \\
 &= p(a) \log \left(\frac{q(t)}{\tilde{q}(t)} \right) + \left(\sum_{a'} p(a') q(t|a') \right) \left(\frac{p(a)}{q(t)} - \frac{\tilde{p}(a)}{\tilde{q}(t)} \right) \\
 &= p(a) \log \left(\frac{q(t)}{\tilde{q}(t)} \right) + q(t) \left(\frac{p(a)}{q(t)} - \frac{\tilde{p}(a)}{\tilde{q}(t)} \right) \\
 &= p(a) \log \left(\frac{q(t)}{\tilde{q}(t)} \right) + p(a) - \frac{q(t)}{\tilde{q}(t)} \tilde{p}(a).
 \end{aligned}$$

Therefore

$$\partial_{q(t|a)} \mathcal{L}_{\beta, \mu}(q(T|A)) = p(a) \log \left(\frac{q(t|a)}{q(t)} \right) - \beta \left(p(a) \log \left(\frac{q(t)}{\tilde{q}(t)} \right) + p(a) - \frac{q(t)}{\tilde{q}(t)} \tilde{p}(a) \right) + \mu_a.$$

Now, recall that here the input set of $q(T|A) = q_{\mathcal{S}}(T|A)$ is $\mathcal{S} = \text{supp}(p(A))$. Hence, we can absorb $p(a)$ into the constant μ_a , and get that a necessary condition for local minimisers of the DIB Lagrangian $\mathcal{L}_{\beta}$ on $\mathcal{Q}_+$ is the existence of constants $(\mu_a)_a \in \mathbb{R}^{|\mathcal{S}|}$ such that

$$\log \left(\frac{q(t|a)}{q(t)} \right) - \beta \left(\log \left(\frac{q(t)}{\tilde{q}(t)} \right) + 1 - \frac{q(t) \tilde{p}(a)}{\tilde{q}(t) p(a)} \right) + \mu_a = 0$$

i.e., such that

$$q(t|a) = q(t) \exp \left[\beta \left(\log \left(\frac{q(t)}{\tilde{q}(t)} \right) + 1 - \frac{q(t) \tilde{p}(a)}{\tilde{q}(t) p(a)} \right) + \mu_a \right].$$

Thus we proved that local minimisers of the DIB Lagrangian $\mathcal{L}_\beta$ over the set of channels $q(T|A) \in C(\mathcal{S}, \mathcal{T})$ with strictly positive entries satisfy the necessary condition

$$q(t|a) = \frac{1}{Z(a, \beta)} q(t) \exp \left[-\beta \left(\frac{q(t)\tilde{p}(a)}{\tilde{q}(t)p(a)} - \log \left(\frac{q(t)\tilde{p}(a)}{\tilde{q}(t)p(a)} \right) - 1 \right) \right], \quad (43)$$

where $Z(a, \beta)$ is a positive normaliser. Note that in (43), we added the factor $\frac{\tilde{p}(a)}{p(a)}$ in the logarithm. This equivalent reformulation is more suited to the implementation of the Blahut-Arimoto algorithm described below. Indeed, in this form, the expression in the exponential is always non-positive (as shown by the study of the function $x \mapsto x - \log(x) - 1$), which avoids overflow for large β .

Note that *a priori*, there might also be local minimisers of $\mathcal{L}_\beta$ on the border of $C(\mathcal{S}, \mathcal{T})$. For the sake of completeness, let us outline an argument showing that this is actually not the case. The computations above show that, deriving $\mathcal{L}_\beta(q(T|A))$ as a function on $\mathcal{Q}_+$, we get

$$\partial_{q(t|a)} \mathcal{L}_\beta(q(T|A)) = p(a) \left[\log \left(\frac{q(t|a)}{q(t)} \right) - \beta \left(\log \left(\frac{q(t)}{\tilde{q}(t)} \right) + 1 - \frac{q(t)\tilde{p}(a)}{\tilde{q}(t)p(a)} \right) \right].$$

In particular, for $q(T|A) \in C(\mathcal{S}, \mathcal{T})$ strictly positive but with at least one coordinate approaching 0, the directional derivative w.r.t this coordinate diverges to $-\infty$. Indeed, $D(p||\tilde{p}) < +\infty$ implies $\mathcal{S} = \text{supp}(p(A)) \subseteq \text{supp}(\tilde{p}(A))$, while each $q(T|a)$ is a probability, with $p(A)$ and $\tilde{p}(A)$ fixed; so that there are strictly positive constants k and K such that $k \leq q(t) := \sum_a p(a)q(t|a) \leq K$ and $k \leq \tilde{q}(t) := \sum_a \tilde{p}(a)q(t|a) \leq K$ for all $t \in \text{supp}(q(T))$ and all $q(T|A) \in \mathcal{Q}_+$. Thus the term $\log \left(\frac{q(t)}{\tilde{q}(t)} \right) + 1 - \frac{q(t)\tilde{p}(a)}{\tilde{q}(t)p(a)}$ remains bounded as well. But as $q(t) \geq k$, on the other hand $\log \left(\frac{q(t|a)}{q(t)} \right)$ diverges to $-\infty$ when $q(t|a)$ goes to 0.

Using classic arguments, we can then use the divergence to $-\infty$ of the gradient close to the border, along with the continuity of $\mathcal{L}_\beta$ on the whole closed set $C(\mathcal{S}, \mathcal{T})$, to prove that $q(T|A)$ cannot be a local minimum of $\mathcal{L}_\beta$ over $C(\mathcal{S}, \mathcal{T})$ if it has a coordinate $q(t|a)$ equal to 0, i.e., if it is on the border of $C(\mathcal{S}, \mathcal{T})$.

D.2.2. BLAHUT-ARIMOTO ALGORITHM

Here, we denote by $C_+(\mathcal{S}, \mathcal{T})$ the subset of $C(\mathcal{S}, \mathcal{T})$ made of channels with only positive entries, by $\Delta_{\mathcal{T},+}$ the open simplex of full-support probabilities on $\mathcal{T}$, and by $\mathbb{R}_+$ the positive real numbers. We define, for $q(T|A) \in C(\mathcal{S}, \mathcal{T})$, a probability $r(T) \in \Delta_{\mathcal{T}}$ on $\mathcal{T}$, and some $m(T) \in (\mathbb{R}_+)^{|\mathcal{T}|}$,

$$\begin{aligned} F(q(T|A), r(T), m(T)) \\ := \sum_{a,t} p(a)q(t|a) \log \left(\frac{q(t|a)}{r(t)} \right) - \beta \sum_{a,t} p(a)q(t|a) \left(\log \left(m(t) \frac{\tilde{p}(a)}{p(a)} \right) - m(t) \frac{\tilde{p}(a)}{p(a)} + 1 \right). \end{aligned}$$

The function F is thus defined on the open and convex set

$$\text{Dom}_F := C_+(\mathcal{S}, \mathcal{T}) \times \Delta_{\mathcal{T},+} \times (\mathbb{R}_+)^{|\mathcal{T}|}.$$

The next proposition defines the Blahut-Arimoto (BA) algorithm adapted to our problem, and describes its properties.

Proposition 19 *The function F is convex in each of its coordinates. Moreover, for $q_i(T|A) \in C_+(\mathcal{S}, \mathcal{T})$, defining*

$$\begin{aligned} r_{i+1}(t) &:= \sum_a p(a) q_i(t|a), \\ m_{i+1}(t) &:= \frac{\sum_a p(a) q_i(t|a)}{\sum_a \tilde{p}(a) q_i(t|a)}, \\ q_{i+1}(t|a) &:= \frac{1}{Z(a, \beta)} r_{i+1}(t) \exp \left[-\beta \left(m_{i+1}(t) \frac{\tilde{p}(a)}{p(a)} - \log \left(m_{i+1}(t) \frac{\tilde{p}(a)}{p(a)} \right) - 1 \right) \right], \end{aligned} \quad (44)$$

where $Z(a, \beta)$ is a positive normaliser, we have:

- (i) All quantities in (44) are well-defined, and $(q_{i+1}(T|A), r_{i+1}(T), m_{i+1}(T)) \in \text{Dom}_F$.
- (ii) $F(q_i(T|A), r_i(T), m_i(T)) = \mathcal{L}_\beta(q_i(T|A)) + K$, where the Lagrangian $\mathcal{L}_\beta$ is defined in (33) and K is a constant that does not depend on i .
- (iii) At each update of $q_i(T|A)$, $r_i(T)$ and $m_i(T)$, the function F is minimised w.r.t. the corresponding coordinate. In particular,

$$F(q_{i+1}(T|A), r_{i+1}(T), m_{i+1}(T)) \leq F(q_i(T|A), r_i(T), m_i(T)).$$

Before proving it, let us first draw the consequences of Proposition 19. Define some $q_0(T|A) \in C_+(\mathcal{S}, \mathcal{T})$, and the corresponding sequence $(q_i(T|A), r_i(T), m_i(T))_{i \geq 1}$ from (44). From point (i), the sequence is included in Dom_F , and from point (ii), we have, for all i ,

$$\mathcal{L}_\beta(q_i(T|A)) = F((q_i(T|A), r_i(T), m_i(T))) - K.$$

From point (iii), this yields a non-increasing sequence of images $(\mathcal{L}_\beta(q_i(T|A)))_i$. As $\mathcal{L}_\beta$ is bounded from below, this implies that this sequence converges. Moreover, as the closure $\overline{C_+(\mathcal{S}, \mathcal{T})} = C(\mathcal{S}, \mathcal{T})$ of $C_+(\mathcal{S}, \mathcal{T})$ is compact, we can, up to extracting a subsequence, assume that $(q_i(T|A))_i$ converges to a point $q_*(T|A) \in C(\mathcal{S}, \mathcal{T})$.⁷ From the definition of $(q_i(T|A))_i$ through (44) and from the continuity of this iterative equation, we obtain that the limit $q_*(T|A)$ satisfies the fixed-point equation (43). Hence we proved the claims made the beginning of Appendix D.2 about this BA algorithm. Note that even though F is convex in each coordinate, we did not prove that F is convex as a whole. Thus we cannot apply the classic BA arguments (Yeung, 2008) to prove that the sequence $(\mathcal{L}_\beta(q_i(T|A)))_i$ converges to a global minimum of $\mathcal{L}_\beta$. However, the statements proved here match exactly the corresponding statements proven for the BA algorithm in the classic IB case (Tishby et al., 2000).

Proof [Proof of Proposition 19]

The convexity of F in each coordinate is straightforward. Point (i) comes from the fact that $q(T|A) \in C(\mathcal{S}, \mathcal{T})$, where $\mathcal{S}$ is the support of $p(A)$, which contains that of $\tilde{p}(A)$ (because $D(p||\tilde{p}) < +\infty$). Point (ii) is a direct computation. Let us now prove point (iii).

7. In practice, in numerical implementations, we always observed the convergence of $(q_i(T|A))_i$, without any subsequence extraction.

For fixed $(r(T), m(T))$, we know that the function $F(\cdot, r(T), m(T))$ is convex on $C_+(\mathcal{S}, \mathcal{T})$, so that the minimum is achieved at points $q(T|A)$ such that $\nabla_{q(T|A)} F(q(T|A), r(T), m(T)) = 0$. A direct computation shows that the latter is equivalent to, for all $a \in \mathcal{S}, t \in \mathcal{T}$,

$$q(t|a) = \frac{1}{Z(a, \beta)} r(t) \exp \left[-\beta \left(m(t) \frac{\tilde{p}(a)}{p(a)} - \log \left(m(t) \frac{\tilde{p}(a)}{p(a)} \right) - 1 \right) \right], \quad (45)$$

where $Z(a, \beta)$ is a positive normaliser. Moreover, it is standard (Yeung, 2008) to prove that, for fixed $(q(T|A), m(T)) \in C_+(\mathcal{S}, \mathcal{T}) \times (\mathbb{R}_+)^{|\mathcal{T}|}$, the minimum of F w.r.t to $r(T)$ is achieved for

$$r(T) = q(T) := \sum_a p(a) q(T|a) \quad (46)$$

Eventually, for fixed $(q(T|A), r(T)) \in C_+(\mathcal{S}, \mathcal{T}) \times \Delta_{\mathcal{T},+}$ the minimum of F w.r.t. $m(T)$ is, again by convexity, achieved if and only if the corresponding gradient vanishes. But we have

$$\begin{aligned} \partial_{m(t)} F(q(T|A), r(T), m(T)) &= \sum_a p(a) q(t|a) \left(\frac{1}{m(t)} - \frac{\tilde{p}(a)}{p(a)} \right) \\ &= \frac{q(t)}{m(t)} - \tilde{q}(t), \end{aligned}$$

so that the gradient w.r.t $m(T)$ cancels if and only if for all $t \in \mathcal{T}$,

$$m(t) = \frac{q(t)}{\tilde{q}(t)} = \frac{\sum_a p(a) q(t|a)}{\sum_a \tilde{p}(a) q(t|a)}. \quad (47)$$

This proves point (iii). ■

D.3. Details on effective cardinality

(Zaslavsky and Tishby, 2019) defines a concept of effective cardinality for the Lagrangian formulation of the classic IB. Here, we adapt this concept to the DIB framework in its primal formulation, i.e., problem (3), and in a way which also encompasses the case $\text{supp}(p(A)) \subsetneq \mathcal{A}$. For $\kappa = q(T|A) \in C(\mathcal{A}, \mathcal{T})$, consider the “probabilistic image of $\mathcal{A}$ through κ ”, i.e.,

$$\kappa \cdot \mathcal{A} := \{t \in \mathcal{T} : \exists a \in \mathcal{A} : q(t|a) > 0\}.$$

Note that this definition depends only on κ and not on $p(A)$. We then define the cardinality of κ as $K(\kappa) := |\kappa \cdot \mathcal{A}|$. However, for $\kappa \in \text{DIB}(\lambda)$, the number $K(\kappa)$ does not necessarily carry any meaningful information about the DIB problem itself: e.g., it can be easily verified that composing any $\kappa \in \text{DIB}(\lambda)$ with a congruent channel γ (which can arbitrarily increase the cardinality $K(\kappa)$) still yields a solution $\gamma \circ \kappa \in \text{DIB}(\lambda)$. This motivates the definition of the minimum number of symbols t necessary to describe the output of a bottleneck encoder κ . Formally:

Definition 20 *The effective cardinality of a DIB solution $\kappa \in \text{DIB}(\lambda)$ is*

$$k(\kappa) := \min_{\gamma \in C(\mathcal{T}): \gamma \circ \kappa \in \text{DIB}(\lambda)} K(\gamma \circ \kappa),$$

i.e., it is the minimum bottleneck cardinality obtained from a post-processing of κ that still produces a DIB solution for the same parameter λ .

Let us fix an arbitrary $\kappa = q(T|A) \in \text{DIB}(\lambda)$, write $q(A, T) := p(A)q(T|A)$, and assume first that $p(A)$ is full support. It can be shown, using the log-sum inequality, that $k(\kappa)$ is the cardinality of the partition $\tilde{\mathcal{T}}$ of $\text{supp}(q(T))$ defined by the equivalence relation $t \sim t' \Leftrightarrow q(A|t) = q(A|t')$.

For the non full support case, denote by $\tilde{p}(A)$ the unique distribution satisfying $D(p||\tilde{p}) = D(p||\mathcal{E})$ (Ay et al., 2017), and note that $D(p||\mathcal{E}) < \infty$ implies $\mathcal{S} \subseteq \tilde{\mathcal{S}}$, where $\mathcal{S} := \text{supp}(p(A))$ and $\tilde{\mathcal{S}} := \text{supp}(\tilde{p}(A))$. It can be easily verified that the value of $q(T|a)$ for $a \in \tilde{\mathcal{S}}^c$ affects neither the target nor the constraint function of the DIB problem (3). However, a direct consequence of Proposition 18 is that if $\tilde{\mathcal{S}} \setminus \mathcal{S} \neq \emptyset$, the image $\kappa \cdot \mathcal{A}$ of $\mathcal{A}$ through a solution $\kappa \in \text{DIB}(\lambda)$ must contain at least one symbol $t_0 \notin \kappa \cdot \mathcal{S}$, on which to send the elements of $\tilde{\mathcal{S}} \setminus \mathcal{S}$. Here we denoted by $\kappa \cdot \mathcal{S}$ the “probabilistic image of $\mathcal{S}$ through the channel $\kappa = q(T|A)$ ”, i.e.,

$$\kappa \cdot \mathcal{S} := \{t \in \mathcal{T} : \exists a \in \mathcal{S} : q(t|a) > 0\}.$$

Note that $\kappa \cdot \mathcal{S} = \text{supp}(q(T))$, for $q(T) := \sum_{a \in \mathcal{S}} p(a)q(T|a)$.

It can be easily verified that the previous paragraph implies that in the non full support case, the effective cardinality becomes $|\tilde{\mathcal{T}}| + 1$. Note that this is the situation we encounter in our numerical experiments (Section 4.1).

We use the above to numerically compute the effective cardinality. Note that the choice of the threshold for rounding $|q(t|a) - q(t|a')|$ to 0 is here important. We choose 10^{-3} .

D.4. Computable form of $D^p(\kappa||C_G)$

Here we provide more details on the divergence introduced in Section 3.4, and prove that it can be computed directly as a divergence between two channels. Let $\mathcal{S} := \text{supp}(p(A))$. For two channels κ, ν in either $C(\mathcal{A}, \mathcal{T})$ or $C(\mathcal{S}, \mathcal{T})$, we define their Kullback-Leibler divergence $D^p(\kappa||\nu)$ with respect to $p = p(A)$, as (Ay et al., 2017)

$$D^p(\kappa||\nu) := \sum_{a \in \mathcal{S}} p(a) D(\kappa(T|a)||\nu(T|a)).$$

We also define, for a group G acting on $\mathcal{A}$, the set of input-symmetric channels w.r.t. G , i.e.,

$$C_G := \{\nu \in C(\mathcal{A}, \mathcal{T}) : \forall \Phi \in G, \nu \circ \Phi = \nu\},$$

and the corresponding divergence of some $\kappa \in C(\mathcal{A}, \mathcal{T})$ from C_G with respect to p as (Ay, 2015)

$$D^p(\kappa||C_G) := \min_{\nu \in C_G} D^p(\kappa||\nu).$$

For all purposes relevant to this article's scope, we have $D^p(\kappa||C_G)$ if and only if $\kappa \circ \Phi = \kappa$ for all $\Phi \in G$. More precisely:

Assume first that $p(A)$ is full support. Then, from the continuity of the KL divergence and the fact that C_G is a closed subset of $C(\mathcal{A}, \mathcal{T})$, we have $D(\kappa||C_G) = 0$ if and only if $\kappa \in C_G$, i.e., $\kappa \circ \Phi = \kappa$ for all $\Phi \in G$.

Let us now drop the full support assumption on $p(A)$, but assume instead that (i) the group G leaves $\mathcal{S}$ invariant, and (ii) the channel $\kappa = q(T|A)$ is as $q'(T|A)$ in equation (37), i.e., it sends $\mathcal{S}^c$ on a single symbol outside the image of $\mathcal{S}$ through κ . From point (i), the action of G on $\mathcal{A}$ induces an action on $\mathcal{S}$, and a corresponding set C_{G_S} . Denote by κ_S the restriction of a channel $\kappa \in C(\mathcal{A}, \mathcal{T})$ to $\mathcal{S}$. Using both points (i) and (ii), it can be easily verified that for all $\kappa \in C(\mathcal{A}, \mathcal{T})$, we have $\kappa \in C_G$ if and only if $\kappa_S \in C_{G_S}$, and that $D^p(\kappa||C_G) = D^p(\kappa_S||C_{G_S})$. From that we can conclude, using the full support case described above, that here we also have $D^p(\kappa||C_G)$ if and only if $\kappa \circ \Phi = \kappa$ for all $\Phi \in G$.

Note that points (i) and (ii) are satisfied in our numerical experiments in Section 4.1, and that they are also automatically satisfied if $p(A)$ is full support.

Let us now provide a form of $D^p(\kappa||C_G)$ which is easier to compute.

Proposition 21 *Fix $p(A) \in \Delta_A$, a finite group G acting on $\mathcal{A}$ and leaving $\mathcal{S}$ invariant, and $\kappa = q(T|A) \in C(\mathcal{A}, \mathcal{T})$. Then*

$$D^p(\kappa||C_G) = D^p(\kappa||\kappa_G)$$

where $\kappa_G = q_G(T|A) \in C(\mathcal{S}, \mathcal{T})$ is defined through, for $a \in \mathcal{S}$ and $t \in \mathcal{T}$,

$$q_G(t|a) := q(t|[a]) := \frac{\sum_{a' \in [a]} p(a')q(t|a')}{p([a])},$$

with $[a]$ the orbit of a under G .

Intuitively, κ_G is the average of the channel κ over the group G acting on its input, computed using the distribution p on the input.

Proof It is enough to prove that for all $\nu \in C_G$,

$$D^p(\kappa||\kappa_G) \leq D^p(\kappa||\nu).$$

For $a \in \mathcal{S}$, we have $[a] \subseteq \mathcal{S}$ (because G leaves $\mathcal{S}$ invariant), and $q_G(T|a')$ is well-defined and constant for $a' \in [a]$. Moreover, for $\nu = r(T|A) \in C_G$, it is straightforward to verify that $r(T|a')$ is also constant for $a' \in [a]$, so that

$$\sum_{a' \in [a]} r(t|a')p(a') = r(t|a)p([a]).$$

Thus, for $\nu = r(T|A) \in C_G$, and $a_1, \dots, a_n$ a system of representatives of all the orbits included in $\mathcal{S}$,

$$\begin{aligned}
 D^p(\kappa||\nu) - D^p(\kappa||\kappa_G) &= \sum_{a \in \mathcal{S}, t \in \mathcal{T}} p(a)q(t|a) \log \left(\frac{q_G(t|a)}{r(t|a)} \right) \\
 &= \sum_{i=1}^n \sum_t \log \left(\frac{q_G(t|a_i)}{r(t|a_i)} \right) \sum_{a \in [a_i]} p(a)q(t|a) \\
 &= \sum_{i=1}^n \sum_t \log \left(\frac{\sum_{a \in [a_i]} q(t|a)p(a)}{\sum_{a \in [a_i]} r(t|a)p(a)} \right) \sum_{a \in [a_i]} p(a)q(t|a) \\
 &= D(q_1||q_2) \geq 0,
 \end{aligned}$$

where q_1 and q_2 are distributions defined on $\mathcal{S}/G \times \mathcal{T}$, through

$$\begin{aligned}
 q_1([a_i], t) &= \sum_{a \in [a_i]} p(a)q(t|a), \\
 q_2([a_i], t) &= \sum_{a \in [a_i]} p(a)r(t|a).
 \end{aligned}$$

■